Mitacs Globalink 2026 Last modified: 16 October 2026

Approximate Covering Numbers of Priors for Bayesian Inverse Problems

A research log of a Mitacs Globalink Research Internship, 2026*

Daniel Amaya Zabala
Mitacs Globalink Research Intern

16 October 2026

Abstract

The number of measurements needed for Bayesian recovery in inverse problems is governed by the approximate covering number of the prior (Adcock and Huang, 2025). This document reports the work carried out over the summer of 2026, during a Mitacs Globalink Research Internship, on estimating this quantity. We first extend the isotropic Gaussian bound of Adcock and Huang (2025) to Gaussian priors with arbitrary covariance, where the complexity is governed by an effective dimension. We then treat subgaussian priors, without assuming independent coordinates, by enclosing the mass in an ellipsoid built from the covariance. Next we show that maps with the Johnson–Lindenstrauss property preserve covering numbers in both directions, which led us to study the covering numbers of manifolds with positive reach and their relation to intrinsic dimension. Finally, we compute the complexity of two sparsity-promoting priors, the spike-and-slab prior and the Laplace prior.

*This work was supported by Mitacs through the Mitacs Globalink Research Internship program (2026), and carried out at the Department of Mathematics, Simon Fraser University, under the supervision of Ben Adcock.

"... put the cutting edge of the chisel against the shell and strike hard. If needed, begin again at many different points until the shell cracks—and you are satisfied."

[...]

"I can illustrate the second approach with the same image of a nut to be opened. The first analogy that came to my mind is of immersing the nut in some softening liquid, and why not simply water? From time to time you rub so the liquid penetrates better, and otherwise you let time pass. The shell becomes more flexible through weeks and months—when the time is ripe, hand pressure is enough, the shell opens like a perfectly ripened avocado!"

Alexander Grothendieck, *Récoltes et Semailles*, pp. 552–553
(**?**), in the translation of **?**

Contents

Notation **4**

1 Introduction **5**
1.1 The project brief . . . 5

2 Preliminaries **6**
2.1 Approximate covering numbers . . . 6
2.2 Two tools . . . 6

3 Stage 1: Gaussian priors with general covariance **7**
3.1 A high-probability ellipsoid . . . 7
3.2 Covering the ellipsoid . . . 8
3.3 Effective dimension . . . 12
3.4 Examples . . . 13

4 Stage 2: Subgaussian priors **13**
4.1 Covers with centres outside the support . . . 14
4.2 The covering number of a subgaussian prior . . . 14

5 Stage 3: Covering numbers under Johnson–Lindenstrauss maps **16**

6 Stage 4: Covering numbers of manifolds **17**
6.1 Medial axis, reach and injectivity radius . . . 18
6.2 What the reach controls . . . 19
6.3 Volume comparison . . . 20
6.4 Covering numbers of the manifold . . . 21

7 Stage 5: Sparsity-promoting priors **21**
7.1 The reference case: exactly s nonzero entries . . . 22
7.2 Mixtures with unequal weights . . . 23
7.3 The spike-and-slab prior . . . 25
7.4 The Laplace prior . . . 27

Acknowledgements **28**

Notation

Symbol	Meaning
General	
$\mathbb{R}^n$, $\|\cdot\|$	Ambient space; Euclidean norm
log	Natural logarithm
$B_\eta(x)$, B_2^r	Closed ball of radius η centred at x; closed unit ball of $\mathbb{R}^r$
ω_r	$\mathrm{Vol}_r(B_2^r) = \pi^{r/2}/\Gamma(r/2+1)$
$\mathrm{Vol}_r(\cdot)$	r-dimensional Lebesgue measure
$K + L$	Minkowski sum $\{x + y : x \in K,\ y \in L\}$
Covering numbers	
$N(K, \eta)$	Covering number of K by closed balls of radius η with centres in K
$N^\circ(K, \eta)$	External covering number: the same with centres anywhere in $\mathbb{R}^n$ (Definition 4.1)
$\mathrm{Cov}_{\eta,\delta}(P)$	η, δ-approximate covering number of P (Definition 2.1)
$\eta > 0$, $\delta \in (0,1)$	Scale (radius of the covering balls); mass left uncovered
$N(X, \rho, \varepsilon)$, $D(X, \rho, \varepsilon)$	Covering and packing numbers of a metric space (Section 6.4)
Gaussian priors (Stage 1)	
$\Sigma \succeq 0$, r	Covariance matrix; its rank
$\lambda_1 \geq \cdots \geq \lambda_r > 0$	Nonzero eigenvalues of Σ
U_r, Λ_r	$[u_1 \mid \cdots \mid u_r]$ with $U_r^\top U_r = I_r$; $\mathrm{diag}(\lambda_1, \ldots, \lambda_r)$
$\Sigma^\dagger$, $\Sigma^{\dagger/2}$	Moore–Penrose pseudoinverse $U_r \Lambda_r^{-1} U_r^\top$; its square root $U_r \Lambda_r^{-1/2} U_r^\top$
$\mathrm{ran}(\Sigma)$	Range of Σ, equal to $\mathrm{span}\{u_1, \ldots, u_r\}$
Z, W	Whitened vectors $\Sigma^{\dagger/2} X$ and $U_r^\top Z$ (Lemma 3.1)
R_δ	Radius $\sqrt{r + 2\sqrt{r \log(1/\delta)} + 2\log(1/\delta)}$
$\mathcal{E}_\delta$, a_j	Concentration ellipsoid; its semi-axes $R_\delta \sqrt{\lambda_j}$ (Corollary 3.2)
E_θ, $b_j(\theta)$	Ellipsoid containing $E + \rho B_2^r$; its semi-axes (Lemma 3.3)
$k(\eta)$	Effective dimension $\#\{j : a_j > \eta/2\}$
Subgaussian priors (Stage 2)	
$\|\cdot\|_{\psi_2}$	Subgaussian norm
K	Subgaussian constant in Eq. (4.1)
R_δ^{sg}, $\mathcal{E}_\delta^{\mathrm{sg}}$	Subgaussian concentration radius and ellipsoid (Lemma 4.6)
Maps (Stage 3)	
$\varepsilon \in (0,1)$	Distortion of a Johnson–Lindenstrauss map (Definition 5.1)
$\kappa_1 \leq \kappa_2$	Lower and upper bi-Lipschitz (or bi-Hölder) constants
Manifolds (Stage 4)	
$\mathcal{M}$, d	Compact smooth d-dimensional submanifold of $\mathbb{R}^n$
$d_{\mathcal{M}}$, $B_r^{\mathcal{M}}(p)$	Geodesic distance; closed geodesic ball
$\mathrm{Med}(A)$, τ_A	Medial axis and reach of a closed set A (Definition 6.1)
Exp_p, $\mathrm{inj}(\mathcal{M})$	Exponential map at p; injectivity radius (Definition 6.2)
II_p, $K(u, v)$	Second fundamental form; sectional curvature
$\mathrm{Cut}(p)$	Cut locus of p; its distance from p is $\mathrm{inj}_p(\mathcal{M})$
K^N, λ	Sectional curvature of N; curvature of the comparison model in Theorem 6.6
$\mathrm{Vol}(\mathcal{M})$	Riemannian volume of $\mathcal{M}$
Sparsity-promoting priors (Stage 5)	
P_S^σ, P_S	$\mathcal{N}(0, \sigma^2 I_S)$, Gaussian on the coordinates in S; $P_S = P_S^1$
$\mathcal{P}_s$	Prior with exactly s nonzero entries (Eq. (7.4))
p_i, P_i, δ_i	Weights, components and uncovered masses in Lemma 7.3
$\mathrm{SS}(p, \sigma^2)$, S	Spike-and-slab prior; its random support
s (Stage 5)	Sparsity level with $\mathbb{P}(\|S\| > s) \leq \delta/2$ (Lemma 7.6)
$R_{\delta'}^{(j)}$	Radius Eq. (7.1)
$\mathrm{Lap}(b)^{\otimes n}$	Product of Laplace distributions with scale b

1 Introduction

In a standard inverse problem, the goal is to recover an unknown vector $x^* \in \mathbb{R}^n$ from noisy linear measurements

$$y = Ax^* + e \in \mathbb{R}^m. \tag{1.1}$$

In the Bayesian approach, x^* is modelled as a draw from a prior distribution $\mathcal{R}$ on $\mathbb{R}^n$, which in practice is replaced by an approximate prior P, for instance one learned from data by a generative model. Adcock and Huang (2025) show that the number of measurements needed for posterior sampling to recover x^* accurately and stably is governed by the *approximate covering number* $\mathrm{Cov}_{\eta,\delta}(P)$ of the prior (Jalal et al., 2021): the smallest number of balls of radius η that cover at least a fraction $1-\delta$ of the mass of P (Definition 2.1). For instance, under the assumptions of Adcock and Huang (2025, Theorem 1.1), with Gaussian noise of level σ and a subgaussian measurement matrix A, a posterior sample $\hat{x}$ satisfies $\|x^* - \hat{x}\| < 34(\eta + \sigma)$ with probability $1 - \mathcal{O}(\delta)$ as soon as

$$m \geq c\left[\log \mathrm{Cov}_{\eta,\delta}(P) + \log(1/\delta)\right], \tag{1.2}$$

where c depends only on the subgaussian parameters of A (Adcock and Huang, 2025, Theorem 1.2). Estimating $\log \mathrm{Cov}_{\eta,\delta}(P)$ for concrete priors is therefore the step that turns the general theory into explicit measurement bounds.

1.1 The project brief

The project was proposed by Ben Adcock as a direct continuation of Adcock and Huang (2025), published at NeurIPS 2025. Its setting is Bayesian inverse problems with learned priors, with a particular focus on generative priors and, specifically, diffusion models. Given y and A as in Eq. (1.1), one recovers x^* by sampling $\hat{x}$ from the approximate posterior $P(\cdot \,|\, y, A)$, and the main results of Adcock and Huang (2025) are nonasymptotic bounds on $\mathbb{P}[\|x^* - \hat{x}\| \geq \epsilon]$. These bounds break down into three components:

(i) closeness of $\mathcal{R}$ and P;
(ii) low complexity of P;
(iii) concentration of the distribution of A.

The project focuses on (ii). Its starting point is the following question from the brief.

> **Question 1.** *How does one estimate (both computationally and theoretically) the complexity of approximate priors, especially generative priors defined by diffusion models?*

This report concerns the theoretical side of Question 1. The work is presented in the order in which it was carried out, in five stages (Sections 3 to 7). A table of notation follows the table of contents.

Remark 1.1 (Tool and computational resource disclosure) Following the Leiden Declaration on Artificial Intelligence and Mathematics (Leiden Declaration, 2026), the author alone conceived the ideas and chose the structure of this text. The author used a large language model (LLM) for the following tasks.

1. *Editing.* The LLM typeset the author's notes with the style file of B. Adcock.
2. *Proofreading.* The LLM checked the author's proofs. It found a false inclusion in the first covering argument of Stage 1 (Example 3.5) and gaps in a first subgaussian bound.

3. *Results.* The LLM help with the ideas and drafting of Lemmas 3.3, 4.3 and 4.6 and Proposition 3.8.

4. *Examples and figures.* The LLM drafted Remarks 3.4 and 4.2 and Examples 3.5, 7.2, 7.4 and 7.5. It computed their numerical values with short Python scripts and drew Figs. 1 and 2 with TikZ.

5. *Sources.* The LLM checked the cited statements against the original references.

The author attributes no authorship or credit to the tool. The author is responsible for the arguments, the results, and the citations.

2 Preliminaries

2.1 Approximate covering numbers

For a set $K \subseteq \mathbb{R}^n$ and $\eta > 0$, we write $N(K, \eta)$ for the classical covering number, the smallest number of closed balls of radius η with centres in K whose union contains K (Vershynin, 2026, Definition 4.2.2).

Definition 2.1 (Approximate covering number; Adcock and Huang, 2025, Definition 2.1)**.** Let P be a probability distribution on $\mathbb{R}^n$ and $\eta, \delta \geq 0$. The η, δ-*approximate covering number* of P is

$$\mathrm{Cov}_{\eta,\delta}(P) = \min \left\{ k \in \mathbb{N} : \exists \{x_i\}_{i=1}^k \subseteq \mathrm{supp}(P),\ P\left(\bigcup_{i=1}^k B_\eta(x_i) \right) \geq 1 - \delta \right\}.$$

When $\delta = 0$ this is the classical covering number of $\mathrm{supp}(P)$ (Adcock and Huang, 2025, Remark 2.2). The following observation is used repeatedly: if $K \subseteq \mathrm{supp}(P)$ satisfies $P(K) \geq 1 - \delta$, then any η-cover of K with centres in K is admissible in Definition 2.1, so

$$\mathrm{Cov}_{\eta,\delta}(P) \leq N(K, \eta). \tag{2.1}$$

Following Block et al. (2022, Definition 2), who speak of the covering number *at scale* ε, we call the radius η the *scale* of the cover. A cover at scale η replaces each covered point by a centre at distance at most η, so it describes the prior up to an error η. In the result of Adcock and Huang (2025, Theorem 1.2) recalled before Eq. (1.2), η also sets the accuracy of recovery, through the error bound $34(\eta + \sigma)$. The covering number grows as the scale decreases. At the other extreme, if η is at least the diameter of $\mathrm{supp}(P)$, then the ball of radius η around any point of $\mathrm{supp}(P)$ contains $\mathrm{supp}(P)$, so $\mathrm{Cov}_{\eta,\delta}(P) = 1$.

Throughout this document we seek *upper* bounds on $\mathrm{Cov}_{\eta,\delta}(P)$. By Eq. (1.2), an upper bound gives a sufficient number of measurements, which is what one needs in order to know what to expect from recovery algorithms. Lower bounds are not pursued.

2.2 Two tools

The first tool bounds covering numbers from above by volumes.

Lemma 2.2 (Volumetric bound; Vershynin, 2026, Proposition 4.2.10)**.** *Let $K \subseteq \mathbb{R}^r$ and $\eta > 0$. Then*

$$N(K, \eta) \leq \frac{\mathrm{Vol}_r\left(K + \frac{\eta}{2} B_2^r\right)}{\mathrm{Vol}_r\left(\frac{\eta}{2} B_2^r\right)}.$$

The centres of the cover produced by the proof of Lemma 2.2 are the points of a maximal η-separated subset of K, so they lie in K, as required by Definition 2.1. Here and below $\mathrm{Vol}_r(B_2^r) = \omega_r = \pi^{r/2}/\Gamma(r/2+1)$.

The second tool is a sharp tail bound for weighted sums of squared Gaussians.

Lemma 2.3 (Laurent and Massart, 2000, Lemma 1)**.** *Let $Y_1, \dots, Y_D$ be i.i.d. $\mathcal{N}(0,1)$ and $b_1, \dots, b_D \geq 0$. Set $\|b\|_\infty = \max_i b_i$ and $\|b\|_2^2 = \sum_i b_i^2$. Then, for every $x > 0$,*

$$\mathbb{P}\left(\sum_{i=1}^{D} b_i(Y_i^2 - 1) \geq 2\|b\|_2\sqrt{x} + 2\|b\|_\infty x\right) \leq \mathrm{e}^{-x}.$$

In particular, if $Y \sim \chi^2_D$, then $\mathbb{P}\left(Y \geq D + 2\sqrt{Dx} + 2x\right) \leq \mathrm{e}^{-x}$.

3 Stage 1: Gaussian priors with general covariance

The question at this stage. The project brief (Section 1.1) asks how the complexity of a prior behaves beyond the two examples worked out by Adcock and Huang (2025). Both of their examples rest on Adcock and Huang (2025, Lemma A.2), a bound for the isotropic Gaussian $\mathcal{N}(0, \sigma^2 I_n)$, and the natural target was a subgaussian version of that lemma. Before attacking subgaussian vectors, I wanted to understand the mechanism of the proof in a setting where every quantity can be computed exactly. The first step was therefore to replace $\sigma^2 I_n$ by an arbitrary covariance matrix Σ. This forces two changes to the isotropic argument: the high-probability set becomes an ellipsoid instead of a ball, and its covering number must be computed with the anisotropy in mind.

Setting. Throughout this section $P = \mathcal{N}(0, \Sigma)$ on $\mathbb{R}^n$, where $\Sigma \succeq 0$ has rank $r \in \{1, \dots, n\}$. By the spectral theorem, $\Sigma = \sum_{j=1}^{n} \lambda_j u_j u_j^\top$ with orthonormal eigenvectors $u_1, \dots, u_n$ and eigenvalues $\lambda_1 \geq \cdots \geq \lambda_r > 0 = \lambda_{r+1} = \cdots = \lambda_n$. Discarding the zero eigenvalues gives the spectral decomposition

$$\Sigma = U_r \Lambda_r U_r^\top, \qquad U_r = [u_1 | \cdots | u_r] \in \mathbb{R}^{n \times r}, \qquad \Lambda_r = \mathrm{diag}(\lambda_1, \dots, \lambda_r), \tag{3.1}$$

where the columns of U_r are orthonormal, that is, $U_r^\top U_r = I_r$. This is the only property of U_r that we use. We write

$$\Sigma^\dagger = U_r \Lambda_r^{-1} U_r^\top, \qquad \Sigma^{\dagger/2} = U_r \Lambda_r^{-1/2} U_r^\top$$

for the Moore–Penrose pseudoinverse of Σ and its square root. Note that $\Sigma^{\dagger/2}\Sigma^{\dagger/2} = \Sigma^\dagger$ and $\mathrm{supp}(P) = \mathrm{ran}(\Sigma) = \mathrm{span}\{u_1, \dots, u_r\}$.

3.1 A high-probability ellipsoid

The first step is to whiten X, that is, to transform it into a standard Gaussian vector on the r-dimensional subspace $\mathrm{ran}(\Sigma)$.

Lemma 3.1 (Whitening)**.** *Let $X \sim \mathcal{N}(0, \Sigma)$ and $Z = \Sigma^{\dagger/2} X$. Then $\|Z\|^2 \sim \chi^2_r$ and $\|Z\|^2 = X^\top \Sigma^\dagger X$.*

Proof. Since Z is a linear image of a centred Gaussian vector, it is a centred Gaussian vector, with covariance

$$\mathbb{E}[ZZ^\top] = \Sigma^{\dagger/2}\,\Sigma\,\Sigma^{\dagger/2} = U_r \Lambda_r^{-1/2}(U_r^\top U_r)\Lambda_r(U_r^\top U_r)\Lambda_r^{-1/2} U_r^\top = U_r U_r^\top,$$

where we used $U_r^\top U_r = I_r$ twice. Let $W = U_r^\top Z \in \mathbb{R}^r$. It is a centred Gaussian vector with covariance $U_r^\top (U_r U_r^\top) U_r = I_r$, so $W \sim \mathcal{N}(0, I_r)$, and its coordinates $W_1, \dots, W_r$ are independent standard normal variables. Moreover $Z = U_r(\Lambda_r^{-1/2} U_r^\top X)$ lies in the range of U_r, so $Z = U_r U_r^\top Z = U_r W$ and $\|Z\| = \|W\|$, because U_r has orthonormal columns. Hence $\|Z\|^2 = W_1^2 + \cdots + W_r^2 \sim \chi_r^2$. Finally, $\Sigma^{\dagger/2}$ is symmetric, so $\|Z\|^2 = X^\top \Sigma^{\dagger/2} \Sigma^{\dagger/2} X = X^\top \Sigma^\dagger X$. □

Choosing the radius. By Lemma 3.1, the question of how much mass of X lies in a region of the form $\{x^\top \Sigma^\dagger x \le R^2\}$ becomes a question about the tail of a χ_r^2 variable. Lemma 2.3 with $D = r$ and $x = \log(1/\delta)$ gives

$$\mathbb{P}\left(\|Z\|^2 \ge r + 2\sqrt{r \log(1/\delta)} + 2\log(1/\delta) \right) \le \delta.$$

The threshold on the left is therefore a radius outside of which at most δ of the mass lies, and we define

$$R_\delta = \sqrt{r + 2\sqrt{r \log(1/\delta)} + 2\log(1/\delta)}. \tag{3.2}$$

Corollary 3.2 (Concentration ellipsoid)**.** *Let $X \sim P = \mathcal{N}(0, \Sigma)$ and $\delta \in (0,1)$. The ellipsoid*

$$\mathcal{E}_\delta = \left\{ x \in \operatorname{ran}(\Sigma) : x^\top \Sigma^\dagger x \le R_\delta^2 \right\} \tag{3.3}$$

satisfies $\mathcal{E}_\delta \subseteq \operatorname{supp}(P)$ and $\mathbb{P}(X \in \mathcal{E}_\delta) \ge 1 - \delta$. It is an r-dimensional ellipsoid with principal directions $u_1, \dots, u_r$ and semi-axis lengths

$$a_j = R_\delta \sqrt{\lambda_j}, \qquad j = 1, \dots, r. \tag{3.4}$$

Proof. The inclusion $\mathcal{E}_\delta \subseteq \operatorname{ran}(\Sigma) = \operatorname{supp}(P)$ holds by definition. If $u \perp \operatorname{ran}(\Sigma)$, then $\langle X, u \rangle \sim \mathcal{N}(0, u^\top \Sigma u) = \mathcal{N}(0,0)$, so $X \in \operatorname{ran}(\Sigma)$ almost surely. By Lemma 3.1 and the choice Eq. (3.2), $\mathbb{P}(X \in \mathcal{E}_\delta) = \mathbb{P}(\|Z\|^2 \le R_\delta^2) \ge 1 - \delta$. Finally, writing $x = U_r c$ with $c \in \mathbb{R}^r$, the condition $x^\top \Sigma^\dagger x \le R_\delta^2$ reads $\sum_{j=1}^r c_j^2/\lambda_j \le R_\delta^2$, that is, $\sum_{j=1}^r c_j^2/a_j^2 \le 1$ with a_j as in Eq. (3.4). □

3.2 Covering the ellipsoid

To bound $N(\mathcal{E}_\delta, \eta)$ with the volumetric bound (Lemma 2.2), we need the volume of the Minkowski sum $\mathcal{E}_\delta + \frac{\eta}{2} B_2^r$. The first version of this argument treated this set as the ellipsoid with semi-axes $a_j + \eta/2$. Its volume would then be $\operatorname{Vol}_r(B_2^r) \prod_j (a_j + \eta/2)$, the factors $\operatorname{Vol}_r(B_2^r) = \pi^{r/2}/\Gamma(r/2+1)$ would cancel in the ratio, and the result would be $\prod_j (1 + 2a_j/\eta)$. However, the Minkowski sum of an ellipsoid and a ball is not an ellipsoid, and in general it is not contained in the ellipsoid with semi-axes $a_j + \eta/2$; Example 3.5 gives a counterexample. The following lemma replaces that step: it gives a family of ellipsoids, each of which does contain the Minkowski sum.

Lemma 3.3 (Minkowski sum of an ellipsoid and a ball)**.** *Let $A \in \mathbb{R}^{r \times r}$ be symmetric positive definite, $E = \{x \in \mathbb{R}^r : x^\top A^{-1} x \le 1\}$ and $\rho > 0$. Then, for every $\theta \in (0,1)$,*

$$E + \rho B_2^r \subseteq E_\theta := \left\{ x \in \mathbb{R}^r : x^\top M_\theta^{-1} x \le 1 \right\}, \qquad M_\theta = \frac{A}{\theta} + \frac{\rho^2}{1-\theta} I_r.$$

Proof. Step 1: a variational formula. Let $M \in \mathbb{R}^{r \times r}$ be symmetric positive definite and $x \in \mathbb{R}^r$, and consider the function

$$\varphi_{M,x}(z) = 2\langle z, x \rangle - z^\top M z, \qquad z \in \mathbb{R}^r.$$

It is a quadratic function of z with Hessian $-2M$, which is negative definite, so $\varphi_{M,x}$ is strictly concave. Completing the square,

$$\varphi_{M,x}(z) = x^\top M^{-1}x - (z - M^{-1}x)^\top M\,(z - M^{-1}x),$$

as one checks by expanding the last term, which equals $z^\top Mz - 2\langle z, x\rangle + x^\top M^{-1}x$. Since M is positive definite, the last term is nonnegative and vanishes exactly at $z = M^{-1}x$, which is also the unique solution of $\nabla\varphi_{M,x}(z) = 2x - 2Mz = 0$. Therefore

$$x^\top M^{-1}x = \sup_{z\in\mathbb{R}^r} \left(2\langle z, x\rangle - z^\top Mz\right). \tag{3.5}$$

Step 2: why this helps. We must show that $x^\top M_\theta^{-1}x \le 1$ when $x = e+b$ with $e \in E$ and $\|b\| \le \rho$. The inverse of the sum $M_\theta = A/\theta + \rho^2 I_r/(1-\theta)$ cannot be split into a contribution of A and a contribution of $\rho^2 I_r$. The right-hand side of Eq. (3.5), on the other hand, is linear in M, so it splits.

Step 3: conclusion. Let $x = e + b$ with $e \in E$ and $\|b\| \le \rho$. For every $z \in \mathbb{R}^r$,

$$2\langle z, e+b\rangle - z^\top M_\theta z = \left(2\langle z, e\rangle - z^\top \tfrac{A}{\theta} z\right) + \left(2\langle z, b\rangle - z^\top \tfrac{\rho^2}{1-\theta} z\right).$$

By Eq. (3.5) with $M = A/\theta$, the first bracket is at most $e^\top(A/\theta)^{-1}e = \theta\, e^\top A^{-1}e \le \theta$. By Eq. (3.5) with $M = \rho^2 I_r/(1-\theta)$, the second bracket is at most $\frac{1-\theta}{\rho^2}\|b\|^2 \le 1-\theta$. Hence $2\langle z, x\rangle - z^\top M_\theta z \le 1$ for every z, and taking the supremum over z in Eq. (3.5) with $M = M_\theta$ gives $x^\top M_\theta^{-1}x \le 1$. □

Remark 3.4 (Why a family of ellipsoids, and which θ) Since $E + \rho B_2^r$ is not an ellipsoid, there is no canonical ellipsoid to compare it with. Lemma 3.3 provides one for each $\theta \in (0,1)$, and θ decides how much weight goes to E and how much to the ball. Take $A = \operatorname{diag}(a_1^2, \dots, a_r^2)$. Then E_θ has semi-axes

$$b_j(\theta) = \left(\frac{a_j^2}{\theta} + \frac{\rho^2}{1-\theta}\right)^{1/2}, \qquad j = 1, \dots, r.$$

By the Cauchy–Schwarz inequality,

$$(a_j + \rho)^2 = \left(\sqrt{\theta}\cdot\frac{a_j}{\sqrt{\theta}} + \sqrt{1-\theta}\cdot\frac{\rho}{\sqrt{1-\theta}}\right)^2 \le \big(\theta + (1-\theta)\big)\left(\frac{a_j^2}{\theta} + \frac{\rho^2}{1-\theta}\right) = b_j(\theta)^2,$$

with equality if and only if $\theta = a_j/(a_j+\rho)$. So every E_θ is at least as large as the ellipsoid with semi-axes $a_j + \rho$ in every direction. If all a_j are equal to a, the choice $\theta = a/(a+\rho)$ gives equality in every direction, and E_θ is exactly the ball of radius $a + \rho$. If the a_j differ, a single θ cannot give equality in every direction. We then choose the θ that makes the volume $\operatorname{Vol}_r(E_\theta) = \operatorname{Vol}_r(B_2^r)\prod_j b_j(\theta)$ smallest, which is where the infimum over θ in Theorem 3.6 below comes from. Lemma 3.3 can also be derived from the convexity of the matrix fractional function $f(x, Y) = x^\top Y^{-1}x$ on $\mathbb{R}^r \times \mathbb{S}_{++}^r$, where $\mathbb{S}_{++}^r$ is the set of symmetric positive definite $r \times r$ matrices (Boyd and Vandenberghe, 2004, Section 3.1.7, Example 3.4). Since $f(tx, tY) = tf(x, Y)$ for $t > 0$, convexity gives

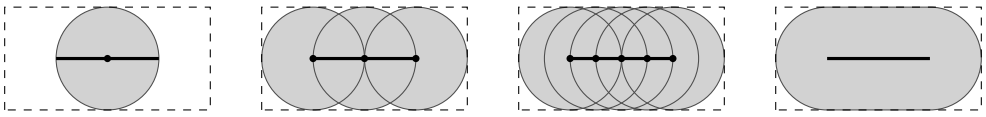
$$f(x_1 + x_2, Y_1 + Y_2) = 2f\left(\tfrac{x_1+x_2}{2}, \tfrac{Y_1+Y_2}{2}\right) \le f(x_1, Y_1) + f(x_2, Y_2).$$

With $x_1 = e$, $Y_1 = A/\theta$, $x_2 = b$ and $Y_2 = \rho^2 I_r/(1-\theta)$, this is the inequality $x^\top M_\theta^{-1}x \le \theta\, e^\top A^{-1}e + \frac{1-\theta}{\rho^2}\|b\|^2$ obtained in Step 3 of the proof.

Example 3.5 (An ellipse that is almost a segment) Let $r = 2$, $a_1 = 1$, $a_2 = \varepsilon$ and $\rho = 1$. As $\varepsilon \to 0$, the ellipse E shrinks to the segment $I = [-1, 1] \times \{0\}$, and we compute the areas in this limit.

The shape of the sum. By definition, $I + B_2^2 = \bigcup_{e \in I} \left(e + B_2^2\right)$ is the union of the unit discs centred at the points of I. Figure 1 builds this union one centre at a time. With one centre, the union is a disc. With three and then five centres, equally spaced on I, the discs overlap and fill a horizontal band. With all the points of I, the union is the set of points at distance at most 1 from I.

- *It lies in a 4×2 rectangle.* A point of the sum has the form $x = (t, 0) + b$ with $|t| \le 1$ and $\|b\| \le 1$, so $|x_1| \le 2$ and $|x_2| \le 1$. Hence $I + B_2^2 \subseteq [-2, 2] \times [-1, 1]$.
- *It does not fill the rectangle.* If $|x_1| \le 1$, the point of I closest to x is $(x_1, 0)$, at distance $|x_2|$. If $x_1 > 1$, the closest point is the endpoint $(1, 0)$, and if $x_1 < -1$, it is $(-1, 0)$. Hence $I + B_2^2$ is the square $[-1, 1] \times [-1, 1]$ together with the two half discs of radius 1 centred at $(\pm 1, 0)$ that lie outside the square. The four corners of the rectangle, such as $(2, 1)$, are not in the sum.
- *Its area.* The square has area 4 and the two half discs together have area π, so the area of the sum is $4 + \pi \approx 7.14$, less than the area 8 of the rectangle.



(a) one centre (b) three centres (c) five centres (d) all of I

Figure 1: The Minkowski sum $I + B_2^2$ of the segment $I = [-1, 1] \times \{0\}$ (thick line) and the unit disc, built one centre at a time. Each centre $e \in I$ (black dot) contributes the disc $e + B_2^2$. As centres are added, the union of the discs fills a horizontal band; with all of I, it is the set of points at distance at most 1 from I. The dashed 4×2 rectangle $[-2, 2] \times [-1, 1]$ contains the sum in every panel, but the sum leaves out its four corners.

The limiting areas of the sum and of three ellipses are as follows.

Set	Semi-axes	Area
$I + \rho B_2^2$	not an ellipse	$4 + \pi \approx 7.14$
Ellipse with semi-axes $a_j + \rho$	$2,\ 1$	$2\pi \approx 6.28$
E_θ, $\theta = 1/2$	$2,\ \sqrt{2}$	$2\sqrt{2}\,\pi \approx 8.89$
E_θ, $\theta = 1/3$	$\sqrt{9/2},\ \sqrt{3/2}$	$\frac{3\sqrt{3}}{2}\pi \approx 8.16$

The ellipse with semi-axes $a_j + \rho$ has *smaller* area than the sum, so it cannot contain it. Concretely, the point $x = (1 + 1/\sqrt{2},\ 1/\sqrt{2})$ belongs to the sum, since it is the point $(1, 0)$ of I plus a vector of length 1. For the ellipse with semi-axes 2 and 1,

$$\frac{x_1^2}{4} + \frac{x_2^2}{1} \approx 0.728 + 0.5 = 1.23 > 1,$$

so x lies outside it, while for $E_{1/2}$, with semi-axes 2 and $\sqrt{2}$,

$$\frac{x_1^2}{4} + \frac{x_2^2}{2} \approx 0.728 + 0.25 = 0.98 \le 1,$$

as Lemma 3.3 guarantees. Figure 2 shows the three ellipses and the point x. In the limit, $b_1(\theta) b_2(\theta) = \left(\theta(1 - \theta)^2\right)^{-1/2}$, which is minimised at $\theta = 1/3$. This explains the last row: the best ellipse of the family has area about 8.16, larger than the area of the sum, as it must be.

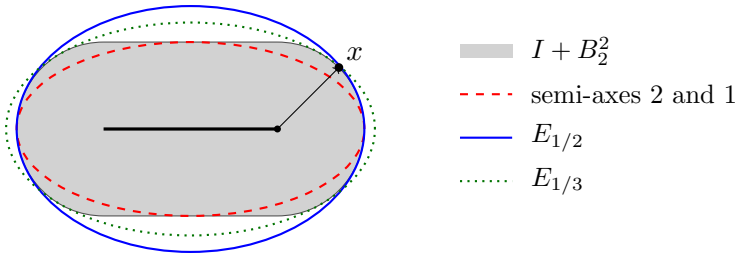


Figure 2: The sum $I + B_2^2$ of Example 3.5 (shaded) and three ellipses. The ellipse with semi-axes $a_j + \rho = 2$ and 1 (dashed) misses part of the sum, for instance the point $x = (1, 0) + (1/\sqrt{2}, 1/\sqrt{2})$. The ellipses $E_{1/2}$ and $E_{1/3}$ of Lemma 3.3 contain the whole sum.

Theorem 3.6 (Approximate covering number of a Gaussian with general covariance)**.** *Let $P = \mathcal{N}(0, \Sigma)$ with Σ as in Eq. (3.1), $\delta \in (0, 1)$ and $\eta > 0$, and let $a_j = R_\delta \sqrt{\lambda_j}$ be as in Eq. (3.4). Then*

$$\mathrm{Cov}_{\eta,\delta}(P) \le \inf_{\theta \in (0,1)} \prod_{j=1}^{r} \left(\frac{1}{\theta} \left(\frac{2a_j}{\eta} \right)^2 + \frac{1}{1-\theta} \right)^{1/2} \le 2^{r/2} \prod_{j=1}^{r} \left(1 + \frac{2a_j}{\eta} \right). \tag{3.6}$$

If all nonzero eigenvalues are equal, $\lambda_1 = \cdots = \lambda_r = \sigma^2$ (for instance $\Sigma = \sigma^2 I_n$, in which case $r = n$), then

$$\mathrm{Cov}_{\eta,\delta}(P) \le \left(1 + \frac{2\sigma R_\delta}{\eta} \right)^r. \tag{3.7}$$

Proof. By Corollary 3.2 and Eq. (2.1), $\mathrm{Cov}_{\eta,\delta}(P) \le N(\mathcal{E}_\delta, \eta)$. The map $c \mapsto U_r c$ is an isometry from $\mathbb{R}^r$ onto $\mathrm{ran}(\Sigma)$ that sends $E = \{c : \sum_j c_j^2/a_j^2 \le 1\}$ onto $\mathcal{E}_\delta$, and covering numbers are preserved by isometries. Hence $N(\mathcal{E}_\delta, \eta) = N(E, \eta)$, where E is the ellipsoid of Lemma 3.3 with $A = \mathrm{diag}(a_1^2, \ldots, a_r^2)$. By Lemmas 2.2 and 3.3 with $\rho = \eta/2$,

$$N(E, \eta) \le \frac{\mathrm{Vol}_r(E_\theta)}{\mathrm{Vol}_r\left(\frac{\eta}{2} B_2^r\right)} = \prod_{j=1}^{r} \frac{\left(a_j^2/\theta + (\eta/2)^2/(1-\theta) \right)^{1/2}}{\eta/2},$$

because E_θ is an ellipsoid with semi-axes $(a_j^2/\theta + \rho^2/(1-\theta))^{1/2}$ and the volume of an ellipsoid is $\mathrm{Vol}_r(B_2^r)$ times the product of its semi-axes. Rearranging and taking the infimum over θ gives the first inequality in Eq. (3.6). The second follows from $\theta = 1/2$ and $\sqrt{2(t^2+1)} \le \sqrt{2}(1+t)$ for $t \ge 0$. If $\lambda_1 = \cdots = \lambda_r = \sigma^2$, then $a_j = \sigma R_\delta$ for all $j \le r$, and the choice $\theta = t/(1+t)$ with $t = 2\sigma R_\delta/\eta$ gives $t^2/\theta + 1/(1-\theta) = (1+t)^2$, which proves Eq. (3.7). □

Remark 3.7 (Comparison with Adcock and Huang, 2025, Lemma A.2) In the isotropic case $\Sigma = \sigma^2 I_n$, the bound in Adcock and Huang (2025, Lemma A.2) reads

$$\mathrm{Cov}_{\eta,\delta}(P) \le \left(1 + \frac{2\sqrt{n}\,\sigma t}{\eta} \right)^n, \quad t = 1 + \sqrt{\tfrac{2}{n} \log(1/\delta)},$$

using a Chernoff bound for the χ_n^2 distribution. The effective radius there is $\sqrt{n}\,t$, and

$$(\sqrt{n}\,t)^2 = \left(\sqrt{n} + \sqrt{2\log(1/\delta)} \right)^2 = n + 2\sqrt{2n\log(1/\delta)} + 2\log(1/\delta).$$

Comparing with Eq. (3.2) (with $r = n$),

$$(\sqrt{n}\,t)^2 - R_\delta^2 = 2(\sqrt{2} - 1)\sqrt{n \log(1/\delta)} > 0,$$

so $R_\delta < \sqrt{n}\,t$ and Eq. (3.7) is strictly smaller than the bound of Adcock and Huang (2025) for every $\delta \in (0,1)$. The gain affects only the constant in the middle term; both radii are $\sqrt{n}(1+o(1))$ as $n \to \infty$ with δ fixed.

3.3 Effective dimension

The factor $2^{r/2}$ in Eq. (3.6) charges every one of the r directions, including those in which the ellipsoid is much thinner than the scale η. Through Theorem 3.6, $\log \mathrm{Cov}_{\eta,\delta}(P)$ is therefore always at least $\frac{r}{2}\log 2$, even when most factors $1 + 2a_j/\eta$ are close to 1. The idea of this subsection is that a direction thinner than $\eta/2$ does not need to be covered at all: think of a pancake, which can be covered by covering its shadow on the long directions, because every ball of radius η already reaches across the short ones.

Proposition 3.8 (Effective dimension bound)**.** *In the setting of Theorem 3.6, let $k \in \{0, 1, \dots, r\}$ be such that $a_{k+1} \le \eta/2$, where $a_{r+1} := 0$. Then*

$$\mathrm{Cov}_{\eta,\delta}(P) \le 2^{k/2} \prod_{j=1}^{k} \left(1 + \frac{4a_j}{\eta}\right). \tag{3.8}$$

In particular, this holds with $k = k(\eta) := \#\{j : a_j > \eta/2\}$, the number of directions in which $\mathcal{E}_\delta$ is wider than $\eta/2$, and then

$$\log \mathrm{Cov}_{\eta,\delta}(P) \le k(\eta) \left(\tfrac{1}{2}\log 2 + \log\left(1 + \frac{4a_1}{\eta}\right)\right).$$

Proof. As in the proof of Theorem 3.6, it suffices to bound $N(E, \eta)$, where $E = \{c \in \mathbb{R}^r : \sum_{j=1}^r c_j^2/a_j^2 \le 1\}$ is the ellipsoid $\mathcal{E}_\delta$ written in the coordinates $c = U_r^\top x$, and to check that the centres lie in $\mathcal{E}_\delta$. Recall that $a_1 \ge a_2 \ge \cdots \ge a_r$.

Step 1: split into head and tail. Write $c = (c', c'')$, where the head $c' = (c_1, \dots, c_k)$ collects the k longest directions and the tail $c'' = (c_{k+1}, \dots, c_r)$ the others. Let

$$E' = \left\{c' \in \mathbb{R}^k : \sum_{j \le k} \frac{c_j^2}{a_j^2} \le 1\right\}$$

be the *shadow* of E on the head directions. If $c \in E$, then $c' \in E'$, since dropping the tail terms only decreases the sum.

Step 2: the tail is short. Let $c \in E$. Since $a_j \le a_{k+1}$ for $j > k$, we have $1 \le a_{k+1}^2/a_j^2$ for those j, so

$$\|c''\|^2 = \sum_{j>k} c_j^2 \le a_{k+1}^2 \sum_{j>k} \frac{c_j^2}{a_j^2} \le a_{k+1}^2 \sum_{j=1}^{r} \frac{c_j^2}{a_j^2} \le a_{k+1}^2 \le \frac{\eta^2}{4}.$$

Thus every point of E lies within distance $\eta/2$ of the flat ellipsoid $E' \times \{0\}$.

Step 3: cover the head. Let $c'_1, \dots, c'_m \in E'$ be the centres of an $(\eta/2)$-cover of E', with $m = N(E', \eta/2)$, and set $c_i = (c'_i, 0)$. Each c_i belongs to E, so the centres $U_r c_i$ lie in $\mathcal{E}_\delta \subseteq \mathrm{supp}(P)$. For $c \in E$, choose i with $\|c' - c'_i\| \le \eta/2$. By Step 2,

$$\|c - c_i\|^2 = \|c' - c'_i\|^2 + \|c''\|^2 \le \frac{\eta^2}{4} + \frac{\eta^2}{4} \le \eta^2.$$

Hence $N(E, \eta) \le N(E', \eta/2)$.

Step 4: count. E' is a k-dimensional ellipsoid with semi-axes $a_1, \dots, a_k$. The proof of Theorem 3.6, with $\theta = 1/2$, applied in $\mathbb{R}^k$ at scale $\eta/2$, gives $N(E', \eta/2) \le 2^{k/2} \prod_{j \le k}(1 + 4a_j/\eta)$, which is Eq. (3.8). The final statement follows from $a_j \le a_1$. □

Remark 3.9 (The role of the effective dimension) The effective dimension $k(\eta)$ plays for Gaussian priors the role that the latent dimension plays in Adcock and Huang (2025, Proposition 4.1): it counts the directions u_j in which the concentration ellipsoid $\mathcal{E}_\delta$ has semi-axis $a_j > \eta/2$. It depends on the scale: as η decreases, more directions satisfy this condition. For priors concentrated near a manifold, the intrinsic dimension plays the same role (Section 6).

3.4 Examples

Example 3.10 (Isotropic covariance) Let $\Sigma = \sigma^2 I_n$. Then $r = n$, $a_j = \sigma R_\delta$ for all j, and Theorem 3.6 gives

$$\mathrm{Cov}_{\eta,\delta}(P) \le \left(1 + \frac{2\sigma R_\delta}{\eta}\right)^n, \qquad R_\delta = \sqrt{n + 2\sqrt{n\log(1/\delta)} + 2\log(1/\delta)},$$

which improves Adcock and Huang (2025, Lemma A.2) by Remark 3.7.

Example 3.11 (Polynomial decay) Let $\lambda_j = j^{-\alpha}$, $j = 1, \dots, n$, for some $\alpha > 0$, so that $r = n$ and $a_j = R_\delta j^{-\alpha/2}$. The condition $a_j > \eta/2$ is equivalent to $j < (2R_\delta/\eta)^{2/\alpha}$, hence

$$k(\eta) \le \min\left\{n, \left(\frac{2R_\delta}{\eta}\right)^{2/\alpha}\right\}, \qquad \log \mathrm{Cov}_{\eta,\delta}(P) \le k(\eta)\left(\tfrac{1}{2}\log 2 + \log\left(1 + \frac{4R_\delta}{\eta}\right)\right)$$

by Proposition 3.8. For fixed η and δ, and since $R_\delta \le \sqrt{n} + \sqrt{2\log(1/\delta)}$, the exponent grows like $n^{1/\alpha}$ up to logarithmic factors, which is sublinear in n when $\alpha > 1$.

Example 3.12 (Exponential decay) Let $\lambda_j = \mathrm{e}^{-\beta j}$, $j = 1, \dots, n$, for some $\beta > 0$, so that $a_j = R_\delta \mathrm{e}^{-\beta j/2}$. Then $a_j > \eta/2$ if and only if $j < \frac{2}{\beta}\log(2R_\delta/\eta)$, so

$$k(\eta) \le \min\left\{n, \frac{2}{\beta}\log\left(\frac{2R_\delta}{\eta}\right)\right\},$$

and $\log \mathrm{Cov}_{\eta,\delta}(P) = \mathcal{O}(\beta^{-1}\log^2(R_\delta/\eta))$. The effective dimension now grows only logarithmically in n (through R_δ) and in $1/\eta$.

What this stage taught us. The proof splits into a *concentration step*, which finds an ellipsoid of mass at least $1 - \delta$ and is the only step that uses the Gaussian distribution, and a *geometric step*, which covers that ellipsoid. Covering the Minkowski sum of an ellipsoid and a ball costs a factor $2^{r/2}$, and directions thinner than $\eta/2$ need no cover, so the complexity is governed by the effective dimension $k(\eta)$ rather than by the rank r. For subgaussian priors, this suggested finding a concentration set of the right shape and reusing the geometric step unchanged.

4 Stage 2: Subgaussian priors

The question at this stage. With the Gaussian case understood, I turned to the original target of the brief: a bound on $\mathrm{Cov}_{\eta,\delta}(P)$ when P is the law of a subgaussian random vector. The plan was to follow Stage 1: find a set that carries mass at least $1 - \delta$, then cover it. A first attempt enclosed the mass in a ball, under the assumption that the coordinates are independent. A ball ignores the shape of the distribution, and Stage 1 had shown that the shape matters. I therefore replaced the ball by an ellipsoid adapted to the covariance, which also removes the independence assumption. This section presents only this general version (Theorem 4.5), which supersedes the first attempt.

We use the subgaussian norm $\|\cdot\|_{\psi_2}$ of Vershynin (2026, Section 2.6). For a random vector, $\|X\|_{\psi_2} = \sup_{\|u\|=1} \|\langle X, u\rangle\|_{\psi_2}$ (Vershynin, 2026, Definition 3.4.1).

4.1 Covers with centres outside the support

Definition 4.1 (External covering number)**.** Let $K \subseteq \mathbb{R}^n$ and $\rho > 0$. The *external covering number* $N^\circ(K, \rho)$ is the smallest number m for which there are points $z_1, \dots, z_m \in \mathbb{R}^n$ with

$$K \subseteq \bigcup_{i=1}^{m} B_\rho(z_i).$$

The name is that of Vershynin (2026, Exercise 4.25). The only difference with $N(K, \rho)$ is that the centres z_i may lie outside K, so $N^\circ(K, \rho) \le N(K, \rho)$.

Remark 4.2 (Why subgaussian priors need it) Definition 2.1 requires the centres to lie in $\operatorname{supp}(P)$. In Stage 1 this is automatic: the concentration ellipsoid $\mathcal{E}_\delta$ lies in $\operatorname{ran}(\Sigma)$, which is the support of $\mathcal{N}(0, \Sigma)$, so Eq. (2.1) applies to it. A subgaussian prior, in contrast, can be discrete. If X is uniform on $\{-1, 1\}^n$, then $\operatorname{supp}(P)$ is the set of the 2^n vertices of the cube, and the concentration ellipsoid of Section 4.2 is a ball that is not contained in $\operatorname{supp}(P)$. Equation (2.1) then does not apply to the ellipsoid itself. There are two remedies. The first applies Eq. (2.1) to the intersection of the ellipsoid with $\operatorname{supp}(P)$, which has the same mass and lies in the support. The second covers the ellipsoid with centres anywhere in $\mathbb{R}^n$, and then moves each centre into the support, at the cost of doubling the radius (Lemma 4.3). Theorem 4.5 uses the first remedy for its volumetric bounds and the second for its bound through the effective dimension, whose cover is not built from volumes.

Lemma 4.3 (Recentring a cover)**.** *Let P be a distribution on $\mathbb{R}^n$, $\delta \in [0, 1)$, $\eta > 0$, and let $K \subseteq \mathbb{R}^n$ be a Borel set with $P(K) \ge 1 - \delta$. Then*

$$\operatorname{Cov}_{\eta,\delta}(P) \le N^\circ(K, \eta/2).$$

The set K need not be contained in $\operatorname{supp}(P)$.

Proof. Step 1: an external cover. Let $m = N^\circ(K, \eta/2)$, and let $z_1, \dots, z_m \in \mathbb{R}^n$ satisfy $K \subseteq \bigcup_{i=1}^m B_{\eta/2}(z_i)$, as in Definition 4.1.

Step 2: moving the centres into the support. Let I be the set of indices i for which $B_{\eta/2}(z_i) \cap K \cap \operatorname{supp}(P) \neq \emptyset$. For $i \in I$, pick a point x_i in this intersection. Then $x_i \in \operatorname{supp}(P)$. Moreover $B_{\eta/2}(z_i) \subseteq B_\eta(x_i)$, because every $y \in B_{\eta/2}(z_i)$ satisfies $\|y - x_i\| \le \|y - z_i\| + \|z_i - x_i\| \le \eta/2 + \eta/2$.

Step 3: the moved balls still cover. Let $y \in K \cap \operatorname{supp}(P)$. By Step 1, $y \in B_{\eta/2}(z_i)$ for some i. This index belongs to I, and $y \in B_\eta(x_i)$ by Step 2. Hence $K \cap \operatorname{supp}(P) \subseteq \bigcup_{i \in I} B_\eta(x_i)$. Since $P(\operatorname{supp}(P)) = 1$ on $\mathbb{R}^n$,

$$P\Big(\bigcup_{i \in I} B_\eta(x_i)\Big) \ge P\big(K \cap \operatorname{supp}(P)\big) = P(K) \ge 1 - \delta.$$

The centres x_i, $i \in I$, are therefore admissible in Definition 2.1, and $\operatorname{Cov}_{\eta,\delta}(P) \le |I| \le m$. □

Lemma 4.3 makes precise the remark of Adcock and Huang (2025, after Definition 2.1) that requiring the centres to lie in $\operatorname{supp}(P)$ changes η at worst by a factor of $1/2$.

4.2 The covering number of a subgaussian prior

The main result of this stage bounds $\operatorname{Cov}_{\eta,\delta}(P)$ for a subgaussian prior by the same ellipsoid geometry as in Stage 1. We impose the subgaussian condition on the one-dimensional marginals of X, measured relative to the covariance.

Assumption 4.4 (Subgaussian with covariance Σ) X is a mean-zero random vector in $\mathbb{R}^n$ with covariance $\Sigma = \mathbb{E}[XX^\top]$ as in Eq. (3.1), and there is $K \geq 1$ such that

$$\|\langle X, u\rangle\|_{\psi_2} \leq K \|\langle X, u\rangle\|_{L^2} = K \left(u^\top \Sigma u\right)^{1/2}, \qquad \forall u \in \mathbb{R}^n. \tag{4.1}$$

Condition (4.1) is the one used by Vershynin (2026, Theorem 4.7.1) for covariance estimation. It does not require independent coordinates. If the coordinates are independent with $\mathbb{E}X_i^2 = \sigma_i^2$ and $\|X_i\|_{\psi_2} \leq K\sigma_i$, then Eq. (4.1) holds with K replaced by CK, by Vershynin (2026, Proposition 2.7.1). A Gaussian $\mathcal{N}(0, \Sigma)$ satisfies it with the absolute constant $K_0 = \|g\|_{\psi_2}$, $g \sim \mathcal{N}(0, 1)$.

For $\delta \in (0, 1)$ we write

$$R_\delta^{\mathrm{sg}} = CK \left(\sqrt{r} + \sqrt{\log(1/\delta)}\right), \tag{4.2}$$

where C is the absolute constant of Vershynin (2026, Proposition 6.2.1). It plays the role of the radius R_δ of Eq. (3.2).

Theorem 4.5 (Approximate covering number of a subgaussian prior)**.** *Let P satisfy Assumption 4.4, let $\delta \in (0, 1)$, $\eta > 0$, and let $a_j = R_\delta^{\mathrm{sg}} \sqrt{\lambda_j}$ for $j = 1, \ldots, r$. Then*

$$\mathrm{Cov}_{\eta,\delta}(P) \leq 2^{r/2} \prod_{j=1}^{r} \left(1 + \frac{2a_j}{\eta}\right),$$

and, for any $k \in \{0, \ldots, r\}$ with $a_{k+1} \leq \eta/4$ (where $a_{r+1} := 0$),

$$\mathrm{Cov}_{\eta,\delta}(P) \leq 2^{k/2} \prod_{j=1}^{k} \left(1 + \frac{8a_j}{\eta}\right).$$

If $\lambda_1 = \cdots = \lambda_r = \sigma^2$, then $\mathrm{Cov}_{\eta,\delta}(P) \leq \left(1 + 2CK\sigma \left(\sqrt{r} + \sqrt{\log(1/\delta)}\right)/\eta\right)^r$.

The bounds have the form of Theorem 3.6 and Proposition 3.8. The radius R_δ becomes R_δ^{sg}, so the distribution enters only through K, the rank r and δ. The first and the last bounds are those of Theorem 3.6 with this radius. In the bound through the effective dimension, the scale η also becomes $\eta/2$, which is the price of Lemma 4.3. The proof has the two steps of Stage 1.

1. *Concentration step.* Find an ellipsoid that carries mass at least $1 - \delta$ (Lemma 4.6 below). This is the only step that uses the distribution.

2. *Geometric step.* Cover this ellipsoid. Here the arguments of Stage 1 apply verbatim, with the two remedies of Remark 4.2 for the position of the centres.

Lemma 4.6 (Subgaussian concentration ellipsoid)**.** *Let $X \sim P$ satisfy Assumption 4.4 and let $\delta \in (0, 1)$. The ellipsoid*

$$\mathcal{E}_\delta^{\mathrm{sg}} = \left\{x \in \mathrm{ran}(\Sigma) : x^\top \Sigma^\dagger x \leq (R_\delta^{\mathrm{sg}})^2\right\}$$

satisfies $\mathbb{P}(X \in \mathcal{E}_\delta^{\mathrm{sg}}) \geq 1 - \delta$, and it has semi-axes $a_j = R_\delta^{\mathrm{sg}} \sqrt{\lambda_j}$, $j = 1, \ldots, r$.

Proof. If $u \perp \mathrm{ran}(\Sigma)$, then $\mathbb{E}\langle X, u\rangle^2 = u^\top \Sigma u = 0$, so $X \in \mathrm{ran}(\Sigma)$ almost surely. Let $W = \Lambda_r^{-1/2} U_r^\top X \in \mathbb{R}^r$. For $v \in \mathbb{R}^r$ with $\|v\| = 1$, Eq. (4.1) with $u = U_r \Lambda_r^{-1/2} v$ gives

$$\|\langle W, v\rangle\|_{\psi_2} = \|\langle X, u\rangle\|_{\psi_2} \leq K \left(v^\top \Lambda_r^{-1/2} U_r^\top \Sigma U_r \Lambda_r^{-1/2} v\right)^{1/2} = K,$$

so W is a mean-zero subgaussian random vector in $\mathbb{R}^r$ with $\|W\|_{\psi_2} \le K$. By Vershynin (2026, Proposition 6.2.1) with $t = \sqrt{\log(1/\delta)}$,

$$\mathbb{P}\left(\|W\| \ge CK\left(\sqrt{r} + \sqrt{\log(1/\delta)}\right)\right) \le \delta.$$

Finally $\|W\|^2 = X^\top U_r \Lambda_r^{-1} U_r^\top X = X^\top \Sigma^\dagger X$. The semi-axes are computed as in Corollary 3.2. □

The proof of Vershynin (2026, Proposition 6.2.1) uses Gaussian replacement and no independence, which is why Lemma 4.6 holds for dependent coordinates.

Proof of Theorem 4.5. Let $K = \mathcal{E}_\delta^{\mathrm{sg}}$, so that $P(K) \ge 1 - \delta$ by Lemma 4.6.

Step 1: the volumetric bounds. Since $X \in \mathrm{ran}(\Sigma)$ almost surely and $\mathrm{ran}(\Sigma)$ is closed, $\mathrm{supp}(P) \subseteq \mathrm{ran}(\Sigma)$. Let $K' = K \cap \mathrm{supp}(P)$. Then $K' \subseteq \mathrm{supp}(P)$, and $P(K') = P(K) \ge 1 - \delta$ because $P(\mathrm{supp}(P)) = 1$. By Eq. (2.1), $\mathrm{Cov}_{\eta,\delta}(P) \le N(K', \eta)$. Identify $\mathrm{ran}(\Sigma)$ with $\mathbb{R}^r$ through the isometry $c \mapsto U_r c$, as in the proof of Theorem 3.6. By Lemma 2.2 and the inclusion $K' \subseteq K$,

$$N(K', \eta) \le \frac{\mathrm{Vol}_r\left(K' + \frac{\eta}{2}B_2^r\right)}{\mathrm{Vol}_r\left(\frac{\eta}{2}B_2^r\right)} \le \frac{\mathrm{Vol}_r\left(K + \frac{\eta}{2}B_2^r\right)}{\mathrm{Vol}_r\left(\frac{\eta}{2}B_2^r\right)}.$$

The proof of Theorem 3.6 bounds the last ratio for any ellipsoid with semi-axes a_j, using only its geometry. This gives the first bound and the bound for equal eigenvalues.

Step 2: the effective dimension. By Lemma 4.3, $\mathrm{Cov}_{\eta,\delta}(P) \le N^\circ(K, \eta/2) \le N(K, \eta/2)$. The proof of Proposition 3.8 bounds $N(\mathcal{E}, \eta')$ for any ellipsoid $\mathcal{E}$ with semi-axes a_j. Applying it with $\eta' = \eta/2$ gives the second bound. □

5 Stage 3: Covering numbers under Johnson–Lindenstrauss maps

The question at this stage. Generative priors are pushforwards $P = G_\sharp\gamma$ of a latent distribution γ on $\mathbb{R}^k$ through a map $G : \mathbb{R}^k \to \mathbb{R}^n$, and Adcock and Huang (2025, Proposition 4.1) bound their covering numbers when G is Lipschitz. Reading Stanczuk et al. (2023), who show that a trained diffusion model reveals the dimension of the data manifold through its score, suggested a finer picture: a map from a latent space to data passes through an intermediate space whose dimension is the *intrinsic* dimension of the data. We asked whether maps that nearly preserve distances, in the sense of the Johnson–Lindenstrauss (JL) lemma, also preserve covering numbers, so that the complexity could be computed in whichever space is most convenient. The first statement was proved for JL maps. We record it here in the more general bi-Lipschitz form, since the proof uses nothing else.

Throughout this section, $N(X, \eta)$ is the covering number of a set X by closed balls of radius η with centres in X, as in Section 2. The convention that centres lie in the set is essential for the lower bounds below.

Definition 5.1 (Johnson–Lindenstrauss property)**.** Let $X \subseteq \mathbb{R}^n$ and $\varepsilon \in (0, 1)$. A map $f : X \to \mathbb{R}^d$ has the *ε-JL property on X* if

$$(1 - \varepsilon)\|x - y\| \le \|f(x) - f(y)\| \le (1 + \varepsilon)\|x - y\|, \qquad \forall x, y \in X.$$

This is the form in which Vershynin (2026, Theorem 5.3.1) states the JL lemma: for a set of N points, a scaled random projection onto $d \ge C\varepsilon^{-2}\log N$ dimensions has the ε-JL property with high probability. Note that f need not be linear, and it only needs to be defined on X.

Lemma 5.2 (Covering numbers under a bi-Lipschitz map)**.** *Let $X \subseteq \mathbb{R}^n$, let $0 < \kappa_1 \le \kappa_2 < \infty$ and let $f : X \to \mathbb{R}^d$ satisfy*

$$\kappa_1 \|x - y\| \le \|f(x) - f(y)\| \le \kappa_2 \|x - y\|, \qquad \forall x, y \in X.$$

Set $Y = f(X)$. Then, for every $\eta > 0$,

$$N\left(X, \tfrac{\eta}{\kappa_1}\right) \le N(Y, \eta) \le N\left(X, \tfrac{\eta}{\kappa_2}\right).$$

Proof. Covering numbers are nonincreasing in the radius, and both inequalities are trivial if the relevant covering number is infinite.

Upper bound. Let $\{x_1, \dots, x_m\} \subseteq X$ be an (η/κ_2)-cover of X with $m = N(X, \eta/\kappa_2)$. Given $y \in Y$, write $y = f(x)$ with $x \in X$ and choose i with $\|x - x_i\| \le \eta/\kappa_2$. Then $\|y - f(x_i)\| \le \kappa_2 \|x - x_i\| \le \eta$. Hence $\{f(x_1), \dots, f(x_m)\} \subseteq Y$ is an η-cover of Y, and $N(Y, \eta) \le m$.

Lower bound. Let $\{y_1, \dots, y_m\} \subseteq Y$ be an η-cover of Y with $m = N(Y, \eta)$, and choose $x_i \in X$ with $f(x_i) = y_i$; this is possible because the centres lie in Y. Given $x \in X$, choose i with $\|f(x) - y_i\| \le \eta$. Then $\kappa_1 \|x - x_i\| \le \|f(x) - f(x_i)\| \le \eta$, so $\{x_1, \dots, x_m\}$ is an (η/κ_1)-cover of X, and $N(X, \eta/\kappa_1) \le m$. □

Corollary 5.3 (Johnson–Lindenstrauss maps)**.** *If f has the ε-JL property on X, then*

$$N\left(X, \tfrac{\eta}{1-\varepsilon}\right) \le N(f(X), \eta) \le N\left(X, \tfrac{\eta}{1+\varepsilon}\right), \qquad \forall \eta > 0.$$

Corollary 5.4 (Bi-Hölder maps)**.** *If $\kappa_1 \|x - y\|^\alpha \le \|f(x) - f(y)\| \le \kappa_2 \|x - y\|^\beta$ for all $x, y \in X$, with $\alpha, \beta, \kappa_1, \kappa_2 > 0$, then*

$$N\left(X, (\eta/\kappa_1)^{1/\alpha}\right) \le N(f(X), \eta) \le N\left(X, (\eta/\kappa_2)^{1/\beta}\right), \qquad \forall \eta > 0.$$

Proof. Repeat the proof of Lemma 5.2: $\|x - x_i\| \le (\eta/\kappa_2)^{1/\beta}$ implies $\|f(x) - f(x_i)\| \le \eta$, and $\|f(x) - f(x_i)\| \le \eta$ implies $\|x - x_i\| \le (\eta/\kappa_1)^{1/\alpha}$. □

What this stage taught us. Distance-preserving maps transport covering numbers in both directions, so the complexity of a generative prior can be computed wherever the geometry is simplest, for instance on the data manifold instead of the latent space. This immediately raised the next question: how do we know, estimate and control the covering number of a manifold?

6 Stage 4: Covering numbers of manifolds

The question at this stage. Stage 3 showed that maps with the Johnson–Lindenstrauss property transport covering numbers. The complexity of a generative prior can therefore be computed on the set where the data lie. Under the manifold hypothesis, this set is a d-dimensional submanifold $\mathcal{M} \subset \mathbb{R}^n$ with $d \ll n$ (Narayanan and Mitter, 2010; Block et al., 2022). The question became: how do we know, estimate and control the covering number of a manifold? This led us to Block et al. (2022), who estimate the intrinsic dimension of a distribution from samples using the Wasserstein distance. Their analysis rests on two-sided bounds for the covering numbers of manifolds with positive reach (Block et al., 2022, Section 2.1 and Appendix A).

My task. My task was to read the geometric part of Block et al. (2022), to ground its concepts (medial axis, reach, injectivity radius, curvature and volume comparison), and to present them to the research group. This section is the written version of that presentation, including the examples I worked out for it.

Standing assumptions and notation. Throughout this section, $\mathcal{M} \subset \mathbb{R}^n$ is a compact, connected, smooth, d-dimensional embedded submanifold without boundary, $1 \leq d \leq n$, with the Riemannian metric induced by $\mathbb{R}^n$. The *geodesic distance* $d_{\mathcal{M}}(p, q)$ is the infimum of the lengths of piecewise smooth curves in $\mathcal{M}$ from p to q (Block et al., 2022, Definition 6). Since segments are shortest paths in $\mathbb{R}^n$,

$$\|p - q\| \leq d_{\mathcal{M}}(p, q), \qquad p, q \in \mathcal{M}. \tag{6.1}$$

We write $\mathrm{Vol}(\mathcal{M})$ for the Riemannian volume of $\mathcal{M}$, which coincides with its d-dimensional Hausdorff measure (Block et al., 2022, Section 2.1), and $\omega_d = \pi^{d/2}/\Gamma(d/2+1)$ for the volume of the unit ball of $\mathbb{R}^d$. We use the Hopf–Rinow theorem (Lee, 2018, Chapter 6): for a *connected* Riemannian manifold, metric completeness with respect to the geodesic distance is equivalent to geodesic completeness, that is, to every geodesic being defined on all of $\mathbb{R}$. Since $\mathcal{M}$ is compact, it is complete as a metric space, and therefore geodesically complete.

6.1 Medial axis, reach and injectivity radius

The reach measures how far $\mathcal{M}$ is from intersecting itself, as seen from the ambient space.

Definition 6.1 (Medial axis and reach; Federer, 1959, Block et al., 2022, Definition 7)**.** Let $A \subseteq \mathbb{R}^n$ be closed and $\mathrm{dist}(z, A) = \min_{a \in A} \|z - a\|$. The *medial axis* of A is the set of points with at least two nearest points in A,

$$\mathrm{Med}(A) = \{z \in \mathbb{R}^n : \exists\, p \neq q \in A \text{ such that } \|z - p\| = \|z - q\| = \mathrm{dist}(z, A)\}.$$

The *reach* of A is

$$\tau_A = \inf_{z \in \mathrm{Med}(A)} \mathrm{dist}(z, A) \in [0, +\infty],$$

with $\tau_A = +\infty$ if $\mathrm{Med}(A) = \emptyset$. We write $\tau = \tau_{\mathcal{M}}$.

The injectivity radius measures the same property from inside $\mathcal{M}$. For $p \in \mathcal{M}$ and $v \in T_p\mathcal{M}$, let γ_v be the geodesic with $\gamma_v(0) = p$ and $\gamma_v'(0) = v$. Since $\mathcal{M}$ is geodesically complete, γ_v is defined on $\mathbb{R}$, and the *exponential map* $\mathrm{Exp}_p : T_p\mathcal{M} \to \mathcal{M}$ is $\mathrm{Exp}_p(v) = \gamma_v(1)$ (see Lee, 2018, Chapters 6 and 10).

Definition 6.2 (Injectivity radius)**.** For $p \in \mathcal{M}$, the injectivity radius $\mathrm{inj}_p(\mathcal{M})$ is the supremum of all $a > 0$ such that Exp_p is a diffeomorphism from the open ball $B_a(0) \subseteq T_p\mathcal{M}$ onto its image. The injectivity radius of $\mathcal{M}$ is

$$\mathrm{inj}(\mathcal{M}) = \inf_{p \in \mathcal{M}} \mathrm{inj}_p(\mathcal{M}).$$

The two quantities depend on different structures. The injectivity radius depends only on the Riemannian metric of $\mathcal{M}$, so it is intrinsic. The reach depends on the embedding $\mathcal{M} \subset \mathbb{R}^n$, so it is extrinsic, and it can be made smaller by changing the embedding (Block et al., 2022, Section 1). The following two examples compute both quantities in simple cases.

Example 6.3 (An affine plane) Let $\mathcal{M} = \{(x_1, x_2, 0) : x_1, x_2 \in \mathbb{R}\} \subset \mathbb{R}^3$, with its induced metric. This $\mathcal{M}$ is not compact, so it lies outside the standing assumptions, but both definitions apply to it.

(i) The geodesics of $\mathcal{M}$ are straight lines, so $\mathrm{Exp}_p(v) = p + v$ for all $v \in T_p\mathcal{M} \equiv \mathbb{R}^2 \times \{0\}$. This map is a diffeomorphism of $T_p\mathcal{M}$ onto $\mathcal{M}$, so $\mathrm{inj}_p(\mathcal{M}) = +\infty$ for every p, and $\mathrm{inj}(\mathcal{M}) = +\infty$.

(ii) For $z = (z_1, z_2, z_3) \in \mathbb{R}^3$ and $q = (q_1, q_2, 0) \in \mathcal{M}$, we have $\|z - q\|^2 = (z_1 - q_1)^2 + (z_2 - q_2)^2 + z_3^2$. This quantity has the unique minimiser $q = (z_1, z_2, 0)$. Every point of $\mathbb{R}^3$ therefore has exactly one nearest point in $\mathcal{M}$, so $\mathrm{Med}(\mathcal{M}) = \emptyset$ and $\tau_{\mathcal{M}} = +\infty$.

Example 6.4 (A circle) Let $\mathcal{S} = \{x \in \mathbb{R}^2 : \|x\| = r\}$ be the circle of radius $r > 0$.

(i) Let $p \in \mathcal{S}$ and let v be a unit tangent vector at p. The geodesic γ_v runs along $\mathcal{S}$ at unit speed, so $\mathrm{Exp}_p(tv)$ is the point reached from p after travelling an arc of length t in the direction v. On the interval $(-\pi r, \pi r)$, this map is a diffeomorphism onto $\mathcal{S} \setminus \{-p\}$. On any larger interval it is not injective, because $\mathrm{Exp}_p(tv) = \mathrm{Exp}_p((t - 2\pi r)v)$. Hence $\mathrm{inj}_p(\mathcal{S}) = \pi r$ for every p, and $\mathrm{inj}(\mathcal{S}) = \pi r$.

(ii) For $z \in \mathbb{R}^2$ and $q \in \mathcal{S}$, we have $\|z - q\|^2 = \|z\|^2 - 2\langle z, q\rangle + r^2$. If $z \neq 0$, the Cauchy–Schwarz inequality shows that this quantity has the unique minimiser $q = rz/\|z\|$. If $z = 0$, every point of $\mathcal{S}$ is at distance r from z. Hence $\mathrm{Med}(\mathcal{S}) = \{0\}$.

(iii) Consequently $\tau_{\mathcal{S}} = \mathrm{dist}(0, \mathcal{S}) = r$.

In particular, $\mathrm{inj}(\mathcal{S}) = \pi\tau_{\mathcal{S}}$.

The reach does not control the injectivity radius from above: Block et al. (2022, Figure 2 and Section 2.1) describe a curve in $\mathbb{R}^2$ with a narrow bottleneck, whose reach is arbitrarily small compared with its injectivity radius.

6.2 What the reach controls

A lower bound on the reach controls the curvature of $\mathcal{M}$, its injectivity radius, and the distortion between the geodesic and the Euclidean distances. We write II_p for the second fundamental form of $\mathcal{M}$ at p and $K(u, v)$ for the sectional curvature of the plane spanned by orthonormal vectors $u, v \in T_p\mathcal{M}$.

Proposition 6.5 (Block et al., 2022, Proposition 8)**.** *Let $\mathcal{M}$ satisfy the standing assumptions, with $\tau > 0$. Then:*

(i) $\|\mathrm{II}_p(u, v)\| \leq 1/\tau$ *for all* $p \in \mathcal{M}$ *and all unit vectors* $u, v \in T_p\mathcal{M}$ *(Niyogi et al., 2008, Proposition 6.1);*

(ii) $\mathrm{inj}(\mathcal{M}) \geq \pi\tau$ *(Aamari et al., 2019, Proposition A.1(ii));*

(iii) $-2/\tau^2 \leq K(u, v) \leq 1/\tau^2$ *for all* $p \in \mathcal{M}$ *and all orthonormal* $u, v \in T_p\mathcal{M}$*;*

(iv) if $p, q \in \mathcal{M}$ *and* $\|p - q\| < 2\tau$*, then* $d_{\mathcal{M}}(p, q) \leq 2\tau \arcsin\big(\|p - q\|/(2\tau)\big)$ *(Boissonnat et al., 2019, Lemma 3).*

Proof. Items (i), (ii) and (iv) are the cited results. For (iii), the Gauss equation (Lee, 2018, Chapter 8) gives

$$K(u, v) = \langle \mathrm{II}_p(u, u), \mathrm{II}_p(v, v)\rangle - \|\mathrm{II}_p(u, v)\|^2.$$

By (i) and the Cauchy–Schwarz inequality, $K(u, v) \leq \|\mathrm{II}_p(u, u)\|\,\|\mathrm{II}_p(v, v)\| \leq 1/\tau^2$ and $K(u, v) \geq -1/\tau^2 - 1/\tau^2 = -2/\tau^2$. □

Example 6.4 shows that the bound (ii) is attained. It also shows that (iv) is attained: if $p, q \in \mathcal{S}$ subtend the angle $\varphi \in [0, \pi]$ at the centre, then $\|p - q\| = 2r \sin(\varphi/2)$ and $d_{\mathcal{S}}(p, q) = r\varphi = 2r \arcsin\big(\|p - q\|/(2r)\big)$, with $r = \tau_{\mathcal{S}}$. In this sense, a circle of radius τ is the most distorted curve allowed by a reach τ (Block et al., 2022, Section 2.1). By Aamari et al. (2019, Theorem 3.4), the reach of $\mathcal{M}$ is small for one of two reasons: either $\mathcal{M}$ has a bottleneck, two distant parts of $\mathcal{M}$ that come close in $\mathbb{R}^n$, or $\mathcal{M}$ bends sharply at some point, where $\|\mathrm{I\!I}_p\|$ equals $1/\tau$.

6.3 Volume comparison

Curvature bounds turn into bounds on the volume of small geodesic balls. We write $B_r^{\mathcal{M}}(p) = \{q \in \mathcal{M} : d_{\mathcal{M}}(p, q) \le r\}$ for the closed geodesic ball, and $\mathrm{Cut}(p)$ for the cut locus of p. The distance from p to $\mathrm{Cut}(p)$ equals the injectivity radius $\mathrm{inj}_p(\mathcal{M})$ of Definition 6.2 (Lee, 2018, Chapter 10). The inequalities we use are the following; we keep the integrand of Gray (2004) and write the volume in our notation (Gray writes $V_m^M(r)$ for the volume of the geodesic ball of radius r centred at m).

Theorem 6.6 (Bishop–Günther inequalities; Gray, 2004, Theorem 3.17)**.** *Let N be a complete d-dimensional Riemannian manifold, let $p \in N$, and let $r > 0$ be no greater than the distance from p to its cut locus. Let K^N denote the sectional curvature of N and let $\lambda \in \mathbb{R}$. Then*

$$K^N \ge \lambda \quad \textit{implies} \quad \mathrm{Vol}_N\left(B_r^N(p)\right) \le \frac{2\pi^{d/2}}{\Gamma(d/2)} \int_0^r \left(\frac{\sin(t\sqrt{\lambda})}{\sqrt{\lambda}}\right)^{d-1} \mathrm{d}t, \tag{6.2}$$

$$K^N \le \lambda \quad \textit{implies} \quad \mathrm{Vol}_N\left(B_r^N(p)\right) \ge \frac{2\pi^{d/2}}{\Gamma(d/2)} \int_0^r \left(\frac{\sin(t\sqrt{\lambda})}{\sqrt{\lambda}}\right)^{d-1} \mathrm{d}t. \tag{6.3}$$

Two remarks make the right-hand sides concrete.

1. The integrand is an entire function of λ, given by the power series

$$\frac{\sin(t\sqrt{\lambda})}{\sqrt{\lambda}} = \sum_{k=0}^{\infty} \frac{(-\lambda)^k\, t^{2k+1}}{(2k+1)!}.$$

   For $\lambda > 0$ it is $\sin(t\sqrt{\lambda})/\sqrt{\lambda}$, for $\lambda = 0$ it is t, and for $\lambda < 0$ it is $\sinh(t\sqrt{-\lambda})/\sqrt{-\lambda}$.

2. *For $\lambda = 0$, the right-hand side is the volume of a Euclidean ball.* At $\lambda = 0$ the integrand is t^{d-1}. Using $\Gamma(x+1) = x\,\Gamma(x)$ with $x = d/2$,

$$\frac{2\pi^{d/2}}{\Gamma(d/2)} \int_0^r t^{d-1}\, \mathrm{d}t = \frac{2\pi^{d/2}}{\Gamma(d/2)} \cdot \frac{r^d}{d} = \frac{\pi^{d/2}}{\frac{d}{2}\,\Gamma\left(\frac{d}{2}\right)} r^d = \frac{\pi^{d/2}}{\Gamma\left(\frac{d}{2}+1\right)} r^d = \omega_d\, r^d,$$

   the volume of a ball of radius r in $\mathbb{R}^d$ (Section 2). Here d is the dimension of N, not the dimension of an ambient space.

We apply Theorem 6.6 to $N = \mathcal{M}$, which is complete. By Proposition 6.5(ii), the distance from any p to its cut locus is at least $\pi\tau$, so the theorem holds for every $r \le \pi\tau$. By Proposition 6.5(iii), we may take $\lambda = -2/\tau^2$ in Eq. (6.2) and $\lambda = 1/\tau^2$ in Eq. (6.3); for $r \le \pi\tau$ the integrand of Eq. (6.3) is nonnegative. Since $\sin(t\sqrt{\lambda})/\sqrt{\lambda} = t + \mathcal{O}(\lambda t^3)$ as $t \to 0$, both right-hand sides are comparable to $\omega_d r^d$ when r is small compared with τ. Thus, below the reach, every geodesic ball of $\mathcal{M}$ has roughly the volume of a Euclidean ball of the same radius in $\mathbb{R}^d$.

6.4 Covering numbers of the manifold

For a metric space (X, ρ) and $\varepsilon > 0$, the covering number $N(X, \rho, \varepsilon)$ is the smallest number of points $x_1, \dots, x_s \in X$ such that the balls of radius ε centred at the x_i cover X, and the packing number $D(X, \rho, \varepsilon)$ is the largest number of points of X at pairwise distance greater than ε (Block et al., 2022, Definition 2). For $X = \mathcal{M}$ and ρ the Euclidean distance, $N(\mathcal{M}, \|\cdot\|, \varepsilon) = N(\mathcal{M}, \varepsilon)$ is the covering number of Section 2.

Lemma 6.7 (Packing and covering; van Handel, 2014, Lemma 5.12, Block et al., 2022, Lemma 3)**.** *For every metric space (X, ρ) and $\varepsilon > 0$,*

$$D(X, \rho, 2\varepsilon) \le N(X, \rho, \varepsilon) \le D(X, \rho, \varepsilon).$$

Proof. A maximal ε-separated subset of X is an ε-net, since a point at distance greater than ε from all its elements could be added to it. A closed ball of radius ε contains at most one point of a 2ε-separated set, by the triangle inequality. □

The main result of Block et al. (2022, Section 2.1) is the following.

Theorem 6.8 (Block et al., 2022, Proposition 9 and Corollary 10)**.** *Let $\mathcal{M} \subset \mathbb{R}^n$ satisfy the standing assumptions, with reach $\tau > 0$. Then, for $0 < \varepsilon < \tau$,*

$$\frac{\mathrm{Vol}(\mathcal{M})}{\omega_d}\, d\, 16^{-d}\, \varepsilon^{-d} \le N(\mathcal{M}, \|\cdot\|, \varepsilon) \le \frac{\mathrm{Vol}(\mathcal{M})}{\omega_d} \left(\frac{\pi}{2}\right)^d \varepsilon^{-d}.$$

The proof in Block et al. (2022, Appendix A) follows four steps. We describe them without tracking constants.

1. By Proposition 6.5(ii) and (iii), the sectional curvatures of $\mathcal{M}$ lie in $[-2/\tau^2, 1/\tau^2]$ and $\mathrm{inj}(\mathcal{M}) \ge \pi\tau$.

2. By Theorem 6.6, the volume of every geodesic ball of radius r small compared with τ lies between two constant multiples of $\omega_d r^d$.

3. *Upper bound.* Take a maximal set of points of $\mathcal{M}$ at pairwise geodesic distance greater than ε. The geodesic balls of radius $\varepsilon/2$ around them are disjoint, so their number is at most $\mathrm{Vol}(\mathcal{M})$ divided by the smallest volume of such a ball. By Lemma 6.7, the same bound holds for the geodesic covering number. *Lower bound.* If geodesic balls of radius ε cover $\mathcal{M}$, their number is at least $\mathrm{Vol}(\mathcal{M})$ divided by the largest volume of such a ball.

4. By Eq. (6.1), a geodesic ε-cover is a Euclidean ε-cover, which transfers the upper bound to $N(\mathcal{M}, \|\cdot\|, \varepsilon)$. By Proposition 6.5(iv), the two distances are comparable below the reach, which transfers the lower bound.

7 Stage 5: Sparsity-promoting priors

The question at this stage. The last task was to compute the approximate covering numbers of two standard sparsity-promoting priors, the spike-and-slab prior and the Laplace prior, as theoretical examples to set next to the generative priors of the previous stages. Under both priors, many entries of x are expected to be zero or small. The reference point is Adcock and Huang (2025, Proposition 4.3), which treats vectors with exactly s nonzero entries.

The idea. A sparse vector carries two pieces of information: *where* its nonzero entries are, that is, its support, and *what* values they take. This suggests covering a sparse prior one support at a time.

1. Fix a support S with $|S| = j$. The vectors with support S lie in a j-dimensional coordinate subspace, and Theorem 3.6 covers a Gaussian on that subspace with about $(C/\eta)^j$ balls.

2. There are $\binom{n}{j}$ supports of size j. The balls that cover the pieces, taken together, cover the union of the pieces, so the numbers of balls add up.

3. Taking logarithms, the total splits into a *support term*, about $\log\binom{n}{j}$, which pays for *where*, and a *value term*, about $j\log(C/\eta)$, which pays for *what*.

Section 7.1 makes this precise when the number of nonzero entries is fixed. Section 7.2 explains what changes when this number is random, and Section 7.3 applies it to the spike-and-slab prior. Section 7.4 treats the Laplace prior, whose samples have no zero entries, so the idea does not apply to it.

Notation. For $S \subseteq \{1, \dots, n\}$ and $\sigma > 0$, we write $P_S^\sigma = \mathcal{N}(0, \sigma^2 I_S)$, where I_S is the diagonal matrix with ones in the positions of S and zeros elsewhere. A vector $x \sim P_S^\sigma$ has independent $\mathcal{N}(0, \sigma^2)$ entries on S and zero entries off S. We write $P_S = P_S^1$. For $j \geq 1$ and $\delta' \in (0, 1)$, let

$$R_{\delta'}^{(j)} = \sqrt{j + 2\sqrt{j\log(1/\delta')} + 2\log(1/\delta')}, \tag{7.1}$$

the radius Eq. (3.2) with $r = j$. All nonzero eigenvalues of $\sigma^2 I_S$ equal σ^2, so Theorem 3.6 gives, for $|S| = j \geq 1$,

$$\mathrm{Cov}_{\eta,\delta'}(P_S^\sigma) \leq \left(1 + \frac{2\sigma R_{\delta'}^{(j)}}{\eta}\right)^j. \tag{7.2}$$

7.1 The reference case: exactly s nonzero entries

Mixtures. Let $\mathcal{I}$ be a finite or countable index set, let P_i, $i \in \mathcal{I}$, be probability distributions on $\mathbb{R}^n$, and let $p_i \geq 0$ satisfy $\sum_{i\in\mathcal{I}} p_i = 1$. The *mixture* with components P_i and weights p_i is the distribution

$$P(A) = \sum_{i\in\mathcal{I}} p_i\, P_i(A), \qquad A \subseteq \mathbb{R}^n \text{ Borel}. \tag{7.3}$$

It is countably additive, because sums of nonnegative terms can be rearranged, and its total mass is $P(\mathbb{R}^n) = \sum_i p_i$. The condition $\sum_i p_i = 1$ is therefore exactly what makes P a probability distribution. A sample of P is drawn in two stages.

1. Draw an index I with $\mathbb{P}(I = i) = p_i$.

2. Given $I = i$, draw X from P_i.

By the law of total probability, $\mathbb{P}(X \in A) = \sum_i \mathbb{P}(I = i)\,\mathbb{P}(X \in A \mid I = i) = \sum_i p_i P_i(A)$, so $X \sim P$. The index I is then forgotten: P is the law of X alone. Every prior of this stage, except the Laplace prior, is a mixture with one component for each support.

Let $s \in \{1, \dots, n\}$. Following Adcock and Huang (2025, Section 4.2), a vector $x \sim \mathcal{P}_s$ is drawn in two steps: choose a support S with $|S| = s$ uniformly at random among the $\binom{n}{s}$ possibilities, and

then draw $x \sim P_S$. In other words, $\mathcal{P}_s$ is the mixture (7.3) with one component P_S of weight $\binom{n}{s}^{-1}$ for each support S with $|S| = s$:

$$\mathcal{P}_s = \binom{n}{s}^{-1} \sum_{|S|=s} P_S. \tag{7.4}$$

Its support is the union of the $\binom{n}{s}$ coordinate subspaces of dimension s. Two facts turn the idea above into a bound. The first is the mixture lemma (Adcock and Huang, 2025, Lemma A.3): if $P = \sum_i p_i P_i$ is a finite mixture, then

$$\mathrm{Cov}_{\eta,\delta}(P) \le \sum_i \mathrm{Cov}_{\eta,\delta}(P_i). \tag{7.5}$$

The second is the bound for one s-dimensional standard Gaussian (Adcock and Huang, 2025, Lemma A.2). Together they give the following result.

Proposition 7.1 (Adcock and Huang, 2025, Proposition 4.3)**.** *For every $\eta > 0$ and $\delta \in (0, 1)$,*

$$\log \mathrm{Cov}_{\eta,\delta}(\mathcal{P}_s) \le s \left[\log\left(\frac{en}{s}\right) + \log\left(1 + \frac{2\sqrt{s}}{\eta}\left(1 + \sqrt{\frac{2}{s}\log(1/\delta)}\right)\right)\right].$$

The two terms are the ones announced above.

- The *support term* $s \log(en/s)$ is an upper bound for $\log \binom{n}{s}$, the logarithm of the number of components. It does not depend on η.
- The *value term* $s \log(1 + \dots)$ is the logarithm of the bound for one s-dimensional Gaussian. It grows like $s \log(1/\eta)$ as $\eta \to 0$.

The slope s in $\log(1/\eta)$ is the dimension of each piece of the support, as d is for a manifold in Theorem 6.8. The factor $\binom{n}{s}$ counts the pieces.

Example 7.2 (Two coordinate axes) Let $n = 2$ and $s = 1$. Then $\mathcal{P}_1 = \frac{1}{2}P_{\{1\}} + \frac{1}{2}P_{\{2\}}$: with probability $\frac{1}{2}$ the sample lies on the horizontal axis, with probability $\frac{1}{2}$ on the vertical axis, and its nonzero entry is $\mathcal{N}(0, 1)$. The support is the cross formed by the two axes. By Eq. (7.5) and Eq. (7.2), we may cover each axis separately:

$$\mathrm{Cov}_{\eta,\delta}(\mathcal{P}_1) \le 2\left(1 + \frac{2R_\delta^{(1)}}{\eta}\right).$$

For the standard Gaussian on $\mathbb{R}^2$, Theorem 3.6 gives $\left(1 + 2R_\delta^{(2)}/\eta\right)^2$ instead. As $\eta \to 0$, the first bound grows like η^{-1} and the second like η^{-2}: the cross is a union of two one-dimensional pieces.

7.2 Mixtures with unequal weights

Why a new lemma is needed. The mixture lemma (7.5) charges every component the same price, whatever its weight. For $\mathcal{P}_s$ this is harmless: all components have the same weight and the same dimension. For the spike-and-slab prior of Section 7.3 it is not. That prior is a mixture over *all* 2^n supports, including the full support $\{1, \dots, n\}$, whose component is a Gaussian on all of $\mathbb{R}^n$. The right-hand side of Eq. (7.5) contains the covering number of this component, which is of order $(1 + 2\sigma R_\delta^{(n)}/\eta)^n$ by Eq. (7.2). The resulting bound is no better than the bound for a

Gaussian on all of $\mathbb{R}^n$, and the sparsity brings no gain. But the components with many nonzero entries have tiny weights. We are allowed to leave a mass δ uncovered, and we can spend part of this budget on discarding the rare components altogether. The following lemma, which I proposed at this stage, makes this precise: each component gets its own uncovered mass δ_i, and components may be discarded.

Lemma 7.3 (Weighted mixture lemma)**.** *Let $P = \sum_{i\in\mathcal{I}} p_i P_i$ be a finite or countable mixture of distributions on $\mathbb{R}^n$, with $p_i > 0$ and $\sum_i p_i = 1$. Let $\mathcal{I}_0 \subseteq \mathcal{I}$ be the set of kept components, and let $\delta_i \in [0,1)$ for $i \in \mathcal{I}_0$ satisfy*

$$\sum_{i\in\mathcal{I}_0} p_i\delta_i + \sum_{i\notin\mathcal{I}_0} p_i \le \delta. \tag{7.6}$$

Then $\mathrm{Cov}_{\eta,\delta}(P) \le \sum_{i\in\mathcal{I}_0} \mathrm{Cov}_{\eta,\delta_i}(P_i)$.

The left-hand side of (7.6) is the total mass left uncovered: the uncovered mass inside the kept components, plus the full mass of the discarded components.

Proof. For $i \in \mathcal{I}_0$, let $C_i \subseteq \mathrm{supp}(P_i)$ be a set of $\mathrm{Cov}_{\eta,\delta_i}(P_i)$ points such that the union U_i of the balls of radius η centred at C_i satisfies $P_i(U_i) \ge 1-\delta_i$. Since $p_i > 0$, we have $\mathrm{supp}(P_i) \subseteq \mathrm{supp}(P)$, so $C = \bigcup_{i\in\mathcal{I}_0} C_i \subseteq \mathrm{supp}(P)$. The union U of the balls of radius η centred at C contains every U_i, hence

$$P(U) \ge \sum_{i\in\mathcal{I}_0} p_i P_i(U_i) \ge \sum_{i\in\mathcal{I}_0} p_i(1-\delta_i) = 1 - \sum_{i\notin\mathcal{I}_0} p_i - \sum_{i\in\mathcal{I}_0} p_i\delta_i \ge 1-\delta,$$

where the equality uses $\sum_i p_i = 1$ and the last inequality is (7.6). □

Taking $\mathcal{I}_0 = \mathcal{I}$ and $\delta_i = \delta$ for all i recovers Eq. (7.5).

Example 7.4 (A rare high-dimensional component) Let $n = 1000$ and $P = 0.99\,P_1 + 0.01\,P_2$, where $P_1 = P_{\{1,\dots,10\}}$ is a standard Gaussian on a 10-dimensional coordinate subspace and $P_2 = \mathcal{N}(0, I_{1000})$. Take $\eta = 1$ and $\delta = 0.02$.

(i) The mixture lemma (7.5) bounds $\mathrm{Cov}_{1,0.02}(P)$ by $\mathrm{Cov}_{1,0.02}(P_1) + \mathrm{Cov}_{1,0.02}(P_2)$. By Eq. (7.2), applied to P_1 with $j = 10$ and to P_2 with $j = 1000$,

$$\mathrm{Cov}_{1,0.02}(P) \le \left(1 + 2R_{0.02}^{(10)}\right)^{10} + \left(1 + 2R_{0.02}^{(1000)}\right)^{1000} \approx \mathrm{e}^{24.9} + \mathrm{e}^{4224.2}.$$

This bound is dominated by the rare component.

(ii) Lemma 7.3 with $\mathcal{I}_0 = \{1\}$ discards P_2. Condition (7.6) reads $0.99\,\delta_1 + 0.01 \le 0.02$, which holds for $\delta_1 = 1/99$. By Eq. (7.2),

$$\log \mathrm{Cov}_{1,0.02}(P) \le \log \mathrm{Cov}_{1,1/99}(P_1) \le 10 \log\left(1 + 2R_{1/99}^{(10)}\right) \approx 25.2.$$

The rare component has weight 0.01, less than $\delta = 0.02$, so it can be left uncovered. The price is a smaller uncovered mass $\delta_1 = 1/99$ for the main component, which changes the radius only slightly, from $R_{0.02}^{(10)} \approx 5.51$ to $R_{1/99}^{(10)} \approx 5.72$.

7.3 The spike-and-slab prior

Let $p \in (0,1]$ and $\sigma > 0$. The *spike-and-slab prior* $P = \mathrm{SS}(p, \sigma^2)$ is the law of

$$X = (B_1 G_1, \ldots, B_n G_n),$$

where $B_1, \ldots, B_n \sim \text{Bernoulli}(p)$ and $G_1, \ldots, G_n \sim \mathcal{N}(0, \sigma^2)$ are all independent. Each entry is exactly zero with probability $1-p$ (the spike at 0) and Gaussian with probability p (the slab). The random support $S = \{i : B_i = 1\}$ has $\mathbb{P}(S = T) = p^{|T|}(1-p)^{n-|T|}$ for every $T \subseteq \{1, \ldots, n\}$, and, conditionally on $S = T$, $X \sim P_T^\sigma$. Thus

$$\mathrm{SS}(p, \sigma^2) = \sum_{T \subseteq \{1,\ldots,n\}} p^{|T|}(1-p)^{n-|T|}\, P_T^\sigma. \tag{7.7}$$

Compared with Eq. (7.4), the number of nonzero entries $|S|$ is now random, with the binomial law $\mathrm{Bin}(n, p)$ and mean np.

Example 7.5 (Spike-and-slab in the plane) Let $n = 2$. By Eq. (7.7),

$$\mathrm{SS}(p, \sigma^2) = (1-p)^2\, \delta_0 + p(1-p)\, P_{\{1\}}^\sigma + p(1-p)\, P_{\{2\}}^\sigma + p^2\, P_{\{1,2\}}^\sigma,$$

where δ_0 is the point mass at the origin. The four components live on the origin, the horizontal axis, the vertical axis and the whole plane, with weights $(1-p)^2$, $p(1-p)$, $p(1-p)$ and p^2. Since $P_{\{1,2\}}^\sigma$ is a Gaussian on $\mathbb{R}^2$, the support of $\mathrm{SS}(p, \sigma^2)$ is all of $\mathbb{R}^2$, and $\mathrm{Cov}_{\eta,0}(\mathrm{SS}(p, \sigma^2)) = +\infty$ for every $\eta > 0$. But the plane carries little mass: if $p = 0.1$, its weight is $p^2 = 0.01$. With $\delta = 0.02$, Lemma 7.3 discards the plane, and the uncovered masses $\delta_i = (\delta - p^2)/(1-p^2)$ for the three other components satisfy (7.6). What remains is a point and two one-dimensional Gaussians, as in Example 7.2. This is what the parameter δ buys: the support is large, but most of the mass lies on low-dimensional pieces.

The strategy. In dimension n, the same reasoning discards every support with too many nonzero entries. We split the budget δ into two halves.

1. *Choose a sparsity level s* such that $\mathbb{P}(|S| > s) \le \delta/2$. The supports with more than s entries carry at most $\delta/2$ of the mass, and we discard them.
2. *Cover each kept support T*, with $|T| \le s$, with uncovered mass $\delta/2$ inside it, using Eq. (7.2).
3. *Count*: there are $\binom{n}{j}$ supports of size j, for $j = 0, \ldots, s$.

The only nonelementary step is the first one. The number of nonzero entries $|S| = B_1 + \cdots + B_n$ is a sum of independent bounded variables, so it concentrates around its mean np, and Bernstein's inequality gives an explicit s.

Lemma 7.6 (Number of nonzero entries)**.** *Let $\delta \in (0,1)$ and*

$$s = \min\left\{n,\ \left\lceil np + \sqrt{2np\log(4/\delta)} + \tfrac{2}{3}\log(4/\delta) \right\rceil\right\}. \tag{7.8}$$

Then $\mathbb{P}(|S| > s) \le \delta/2$.

Proof. If $s = n$ there is nothing to prove, since $|S| \le n$. Otherwise we proceed in four steps.

1. The variables $B_i - p$ are independent, have mean zero and satisfy $|B_i - p| \le 1$. The variance of their sum is $np(1-p) \le np$. Bernstein's inequality for bounded variables (Vershynin, 2026, Theorem 2.9.5) gives, for $t \ge 0$,

$$\mathbb{P}\big(||S| - np| \ge t\big) \le 2\exp\left(-\frac{t^2/2}{np(1-p) + t/3}\right) \le 2\exp\left(-\frac{t^2/2}{np + t/3}\right).$$

2. Let $x = \log(4/\delta)$, so that $2\mathrm{e}^{-x} = \delta/2$. The right-hand side of Step 1 is at most $\delta/2$ as soon as $t^2/2 \ge x(np + t/3)$, that is, as soon as

$$t^2 - \tfrac{2}{3}xt - 2npx \ge 0.$$

3. This quadratic inequality holds for every t at least its positive root $t^* = \frac{x}{3} + \sqrt{\frac{x^2}{9} + 2npx}$. Since $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$ for $a, b \ge 0$, we have $t^* \le \sqrt{2npx} + \frac{2}{3}x$. We take $t = \sqrt{2npx} + \frac{2}{3}x$.

4. By Eq. (7.8), $s \ge np + t$. Hence $|S| > s$ implies $|S| - np > t$, and $\mathbb{P}(|S| > s) \le \delta/2$ by Steps 1 and 2.

□

The three terms of Eq. (7.8) have a clear meaning. The term np is the mean number of nonzero entries. The term $\sqrt{2np\log(4/\delta)}$ is a Gaussian-type fluctuation: it is $\sqrt{2\log(4/\delta)}$ times the bound $\sqrt{np}$ on the standard deviation of $|S|$. The term $\frac{2}{3}\log(4/\delta)$ accounts for the occasional large deviations of a sum of Bernoulli variables, and it matters when np is small. Any other s with $\mathbb{P}(|S| > s) \le \delta/2$ can be used in the next theorem, for instance the exact quantile of the binomial law.

Theorem 7.7 (Approximate covering number of the spike-and-slab prior)**.** *Let $P = \mathrm{SS}(p, \sigma^2)$, $\delta \in (0,1)$ and $\eta > 0$, and let $s \in \{1, \dots, n\}$ satisfy $\mathbb{P}(|S| > s) \le \delta/2$, for instance Eq. (7.8). Then*

$$\log \mathrm{Cov}_{\eta,\delta}(P) \le s\log\left(\frac{\mathrm{e}n}{s}\right) + s\log\left(1 + \frac{2\sigma R^{(s)}_{\delta/2}}{\eta}\right).$$

Proof. We follow the three steps of the strategy.

Step 1: discarding the dense supports. Apply Lemma 7.3 to Eq. (7.7) with $\mathcal{I}_0 = \{T : |T| \le s\}$ and $\delta_T = \delta/2$ for $T \in \mathcal{I}_0$. Condition (7.6) holds, because

$$\sum_{|T| \le s} \mathbb{P}(S = T)\,\frac{\delta}{2} + \mathbb{P}(|S| > s) \le \frac{\delta}{2} + \frac{\delta}{2} = \delta.$$

Step 2: covering each kept support. For $|T| = j \ge 1$, Eq. (7.2) gives $\mathrm{Cov}_{\eta,\delta/2}(P^\sigma_T) \le (1 + 2\sigma R^{(j)}_{\delta/2}/\eta)^j$. For $T = \emptyset$, $P^\sigma_\emptyset = \delta_0$ and one ball suffices.

Step 3: counting. There are $\binom{n}{j}$ supports of size j, and $R^{(j)}_{\delta/2}$ is nondecreasing in j. Hence

$$\mathrm{Cov}_{\eta,\delta}(P) \le \sum_{j=0}^{s}\binom{n}{j}\left(1 + \frac{2\sigma R^{(j)}_{\delta/2}}{\eta}\right)^j \le \left(\sum_{j=0}^{s}\binom{n}{j}\right)\left(1 + \frac{2\sigma R^{(s)}_{\delta/2}}{\eta}\right)^s.$$

Finally, $\sum_{j=0}^{s}\binom{n}{j} \le (\mathrm{e}n/s)^s$ for $1 \le s \le n$ (Vershynin, 2026, Exercise 0.6). □

Theorem 7.7 has the same two terms as Proposition 7.1, with the fixed sparsity replaced by s: a support term $s\log(en/s)$ and a value term that grows like $s\log(1/\eta)$. With s from Eq. (7.8), s is close to np when np is large compared with $\log(4/\delta)$, and the support term is then about $np\log(\mathrm{e}/p)$. Through Eq. (1.2), this gives a sufficient number of measurements of order $np\log(1/p)$, plus the value term.

7.4 The Laplace prior

Let $b > 0$. The *Laplace prior* $P = \mathrm{Lap}(b)^{\otimes n}$ is the law of a vector with independent entries of density $(2b)^{-1}\mathrm{e}^{-|t|/b}$, so P has density

$$f(x) = (2b)^{-n}\exp\left(-\|x\|_1/b\right), \qquad x \in \mathbb{R}^n.$$

It is called sparsity promoting because of its maximum a posteriori (MAP) estimator. If $e \sim \mathcal{N}(0, \sigma_e^2 I_m)$ in Eq. (1.1), the posterior density of x given y is proportional to

$$\exp\left(-\frac{\|y - Ax\|^2}{2\sigma_e^2} - \frac{\|x\|_1}{b}\right),$$

so the MAP estimator is

$$\hat{x}_{\mathrm{MAP}} \in \operatorname*{argmin}_{x\in\mathbb{R}^n} \frac{1}{2\sigma_e^2}\|y - Ax\|^2 + \frac{1}{b}\|x\|_1,$$

the LASSO (Tibshirani, 1996), whose minimisers are typically sparse. The samples of P, on the other hand, are not sparse at all.

- Each entry X_i has a density, so $\mathbb{P}(X_i = 0) = 0$. Almost surely, no entry of a sample is zero.
- For $t > 0$, $\mathbb{P}(|X_i| \le t) = 1 - \mathrm{e}^{-t/b}$. For $n = 1000$ and $t = 0.01\,b$, the expected number of entries with $|X_i| \le t$ is $1000(1 - \mathrm{e}^{-0.01}) \approx 10$. A spike-and-slab sample with $p = 0.01$ has, on average, 990 entries equal to zero.

Since a sample has no zero entries, there are no supports to discard, and the strategy of Section 7.3 does not apply. The next result bounds the approximate covering number of the Laplace prior directly. The bound has the form of the bound of Theorem 3.6 for an isotropic Gaussian on $\mathbb{R}^n$.

Proposition 7.8 (Approximate covering number of the Laplace prior)**.** *Let $P = \mathrm{Lap}(b)^{\otimes n}$, $\delta \in (0,1)$ and $\eta > 0$. Then*

$$\log \mathrm{Cov}_{\eta,\delta}(P) \le n\log\left(1 + \frac{2b\sqrt{n}\log(n/\delta)}{\eta}\right).$$

Proof. For each i, $\mathbb{P}(|X_i| > t) = \mathrm{e}^{-t/b}$. With $t = b\log(n/\delta)$, a union bound over the n entries gives $\mathbb{P}(\|X\|_\infty > t) \le n\mathrm{e}^{-t/b} = \delta$. Since $\{\|x\|_\infty \le t\} \subseteq B_2^n(0, t\sqrt{n})$, the ball $B_2^n(0, t\sqrt{n})$ has mass at least $1 - \delta$, and it lies in $\mathrm{supp}(P) = \mathbb{R}^n$. By Eq. (2.1) and Lemma 2.2, $\mathrm{Cov}_{\eta,\delta}(P) \le (1 + 2t\sqrt{n}/\eta)^n$. □

Example 7.9 (Laplace against spike-and-slab) Let $n = 1000$ and $\delta = 0.01$. We compare the spike-and-slab prior $\mathrm{SS}(p, \sigma^2)$ with $p = 0.01$ and $\sigma = 1$ and the Laplace prior $\mathrm{Lap}(b)^{\otimes n}$ with $b = \sqrt{0.005} \approx 0.0707$. With this choice, $\mathbb{E}X_i^2 = 2b^2 = 0.01$ for the Laplace prior and $\mathbb{E}X_i^2 = p\sigma^2 = 0.01$ for the spike-and-slab prior, so the entries of both priors have the same variance.

- *Spike-and-slab.* On average a sample has $np = 10$ nonzero entries. With $\log(4/\delta) = \log 400 \approx 5.99$, Eq. (7.8) gives

$$s = \left\lceil 10 + \sqrt{20 \times 5.99} + \tfrac{2}{3} \times 5.99 \right\rceil = \lceil 10 + 10.95 + 3.99 \rceil = 25.$$

  In Theorem 7.7, the support term is $25 \log(40\mathrm{e}) \approx 117.2$, and the value term uses $R^{(25)}_{0.005} \approx 7.66$ from Eq. (7.1).

- *Laplace.* In Proposition 7.8, $\log(n/\delta) = \log(10^5) \approx 11.51$, so $2b\sqrt{n}\log(n/\delta) \approx 51.5$.

The two upper bounds on $\log \mathrm{Cov}_{\eta,\delta}$ are as follows.

η	Spike-and-slab (Theorem 7.7)	Laplace (Proposition 7.8)
1	187.0	3960.6
0.1	243.2	6245.9
0.01	300.6	8546.7

As η decreases by a factor 10, the spike-and-slab bound grows by 56.2 and then by 57.4, close to $s \log 10 \approx 57.6$. The Laplace bound grows by 2285.3 and then by 2300.8, close to $n \log 10 \approx 2302.6$. So the slope in $\log(1/\eta)$ is $s = 25$ for the spike-and-slab prior and $n = 1000$ for the Laplace prior. The entries of both priors have the same variance, but only the spike-and-slab samples have zero entries, and the bound of Theorem 7.7 uses them.

What this stage taught us. A sparse prior is a mixture over supports, and its covering number splits into a support term, which pays for where the nonzero entries are, and a value term, which grows like the number of nonzero entries times $\log(1/\eta)$. When this number is random, the weighted mixture lemma spends part of the uncovered mass δ on discarding the rare supports with many nonzero entries. The spike-and-slab bound is then controlled by the expected number of nonzero entries, while the Laplace bound has the form of a bound for a full-dimensional Gaussian, because Laplace samples have no zero entries.

Acknowledgements

I am deeply grateful to my supervisor, Ben Adcock, for his patience, his guidance and his support. He gave me the freedom to follow my own ideas and the guidance to correct them, and every meeting with him left me with a clearer view of the next question.

I thank Nick and Max for their company after the meetings, and for making research mathematics more fun. Those moments made the work lighter and the summer more joyful.

This work was supported by Mitacs through the Mitacs Globalink Research Internship program, and it was carried out at the Department of Mathematics of Simon Fraser University. I thank both for this opportunity.

Last, I thank all the friends who made my days unforgettable: for so many moments and love.

References

Aamari, E., Kim, J., Chazal, F., Michel, B., Rinaldo, A., and Wasserman, L. (2019). Estimating the reach of a manifold. *Electronic Journal of Statistics*, 13(1):1359–1399.

Adcock, B. and Huang, N. (2025). How many measurements are enough? Bayesian recovery in inverse problems with general distributions. In *Advances in Neural Information Processing Systems*, volume 38, pages 97285–97326. 39th Conference on Neural Information Processing Systems (NeurIPS 2025). arXiv:2505.10630.

Block, A., Jia, Z., Polyanskiy, Y., and Rakhlin, A. (2022). Intrinsic dimension estimation using Wasserstein distance. *Journal of Machine Learning Research*, 23(313):1–37.

Boissonnat, J.-D., Lieutier, A., and Wintraecken, M. (2019). The reach, metric distortion, geodesic convexity and the variation of tangent spaces. *Journal of Applied and Computational Topology*, 3:29–58.

Boyd, S. and Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press.

Federer, H. (1959). Curvature measures. *Transactions of the American Mathematical Society*, 93(3):418–491.

Gray, A. (2004). *Tubes*, volume 221 of *Progress in Mathematics*. Birkhäuser, second edition.

Jalal, A., Karmalkar, S., Dimakis, A. G., and Price, E. (2021). Instance-optimal compressed sensing via posterior sampling. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 4709–4720.

Laurent, B. and Massart, P. (2000). Adaptive estimation of a quadratic functional by model selection. *The Annals of Statistics*, 28(5):1302–1338.

Lee, J. M. (2018). *Introduction to Riemannian Manifolds*, volume 176 of *Graduate Texts in Mathematics*. Springer, second edition.

Leiden Declaration (2026). Leiden Declaration on Artificial Intelligence and Mathematics. Public statement, 2 June 2026. https://leidendeclaration.ai.

Narayanan, H. and Mitter, S. (2010). Sample complexity of testing the manifold hypothesis. In *Advances in Neural Information Processing Systems*, volume 23, pages 1786–1794.

Niyogi, P., Smale, S., and Weinberger, S. (2008). Finding the homology of submanifolds with high confidence from random samples. *Discrete & Computational Geometry*, 39(1–3):419–441.

Stanczuk, J., Batzolis, G., Deveney, T., and Schönlieb, C.-B. (2023). Your diffusion model secretly knows the dimension of the data manifold. arXiv preprint arXiv:2212.12611v5.

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B (Methodological)*, 58(1):267–288.

van Handel, R. (2014). Probability in High Dimension. Lecture notes (APC 550), Princeton University.

Vershynin, R. (2026). *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge University Press, second edition. Theorem, proposition and exercise numbers refer to the pre-publication version of 22 July 2026.